

Online Shoppers Purchasing Intention

Final Project

Algoritmi e Applicazioni per l'Intelligenza Artificiale

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Abstract

Predicting real-time purchase intent in e-commerce environments is a critical challenge for dynamic personalization and customer retention. This study proposes an end-to-end machine learning pipeline applied to the *Online Shoppers Purchasing Intention* dataset, comprising 12,330 web sessions. To overcome the severe class imbalance and capture complex behavioral patterns, we design a hybrid framework integrating unsupervised representation learning and robust supervised classification. Our methodology initiates with an exhaustive exploratory data analysis (EDA), including Google Analytics metrics and temporal dynamics. We then apply Principal Component Analysis (PCA) for linear dimensionality reduction and *t*-Distributed Stochastic Neighbor Embedding (*t*-SNE) to map non-linear manifolds. Crucially, we leverage *K*-Means clustering on the latent space to engineer a novel categorical feature, `Cluster_ID`, representing distinct user navigation profiles. To prevent data leakage, a strict stratified train-test split (80/20) is enforced prior to applying Information Theory filters (Mutual Information) and Synthetic Minority Over-sampling Technique for Nominal and Continuous features (SMOTENC) on the training set alone. Finally, we benchmark four distinct learning paradigms: Random Forest, XGBoost, Support Vector Machines (SVM) with radial basis function kernels, and a deep Multi-Layer Perceptron (MLP). In classical ML algorithms, hyperparameters are systematically optimized using Grid Search coupled with a Stratified 5-fold Cross-Validation; the only DL algorithm is improved using stronger regularization and Early Stopping. The models are evaluated under a unified framework prioritizing Precision-Recall curves, F1-Score, and Area Under the ROC (AUC-ROC), providing insights into the trade-off between false alarm rates and missed conversion opportunities.

Contents

1	Introduction	3
1.1	Conversion in E-commerce	3
1.2	Dataset Description	3
1.3	Project Objectives	4
2	Explorational Data Analysis (EDA)	4
2.1	Numerical Features Analysis	4
2.2	Categorical Features Analysis	6
2.3	Target Analysis and Class Imbalance	8
2.4	Time Context and Special Days	9
2.5	Google Analytics Metrics	10
2.6	Multivariate Correlation	11
3	Pre-processing and Dimensionality Reduction	12
3.1	Categorical Encoding and Feature Selection	12
3.2	Feature Scaling	13
3.3	Principal Component Analysis (PCA)	14

3.4	t-SNE Non-linear Exploration	16
4	Clustering and Feature Engineering	18
4.1	Optimal Cluster Number (k)	18
4.2	Clustering Comparison ($k=2, 3, 4$)	19
4.3	Cluster Profiling	20
5	Data Preparation	21
5.1	Stratified Train-Test Split	21
5.2	Mutual Information and Feature Selection	22
5.3	Synthetic Balancing: SMOTE-NC	23
6	Supervised Learning Modeling	24
6.1	Random Forest	25
6.2	XGBoost	29
6.3	Support Vector Machines (SVM)	30
6.4	Multi-Layer Perceptron (MLP)	31
7	Comparisons and Conclusions	33
8	AI Disclosure	36
9	References	36

1 Introduction

1.1 Conversion in E-commerce

In recent years, the spread of e-commerce and the increase in its use by consumers have necessitated a paradigm shift within the retail sector. The expertise and experience of retailers in physical retail must inevitably give way to virtual mechanisms for predicting customer behaviour and personalising the shopping experience, with a view to optimising sales. The development of this sector offers enormous potential, which lies in the ability to learn from the behaviour of online visitors, monitor their decisions and classify each individual's purchasing intentions based on the results obtained.

By working with the *Online Shoppers Purchase Intention* dataset, it is possible to extract value from the available data using various tools and methodologies, with the aim of identifying the common thread that guides visitors to online shopping sites and seeking to convert each session into a potential purchase. The importance of this initiative also lies in the possibility of applying it to different types of customers, tailoring the service to each individual's requirements with the aim of making it as user-friendly as possible.

1.2 Dataset Description

The dataset consists of feature vectors belonging to 12,330 online sessions. The values are derived from the URL information of the pages visited by the user and updated in real time when a user takes an action, e.g. moving from one page to another. The dataset was formed so that each session would belong to a different user in a 1-year period to avoid any tendency to a specific campaign, special day, user profile or period.

Each sample contains values for 18 features, 10 numerical and 8 categorical, grouped into different semantic sections:

1. Browsing Activity – Behavioural Features:

- **Administrative:** Number of 'administrative' pages (e.g. account management, login, checkout) visited by the user during the session.
- **Administrative.Duration:** Total time (in seconds) spent by the user on pages belonging to the administrative category.
- **Informational:** Number of 'informational' pages (e.g. About Us, Contact Us, Privacy Policy) visited during the session.
- **Informational.Duration:** Total time (in seconds) spent on the information pages.
- **ProductRelated:** Number of product pages viewed by the user.
- **ProductRelated.Duration:** Total time spent viewing product pages, which is an indication of the level of commercial interest.

2. Google Analytics metrics – Performance indicators:

- **BounceRates:** This represents the percentage of visitors who enter the site via a specific page and leave it without sending any further requests to the server.
- **ExitRates:** Calculated for each page as the percentage of page views that were the last ones in the session.
- **PageValues:** This indicates the average number of web pages visited before a user completes an e-commerce transaction.

In our case, these features do not refer to specific pages, but are average values relating to the pages in the three categories under consideration.

3. Time Context – Cyclical and seasonal features:

- **SpecialDay:** A numerical value (from 0 to 1) indicating how close the session is to a special event (e.g. Valentine's Day, Mother's Day). The value is weighted according to the probability of purchase, peaking approximately one week before the event.
- **Month:** A categorical variable indicating the month in which the browsing session took place.

4. User Profile and Traffic – Demographic and Technical Features:

- **OperatingSystems**: The numerical identifier of the operating system used by the user.
- **Browser**: The numerical identifier of the browser used for the session.
- **Region**: A numerical code identifying the geographical area from which the user’s connection originates.
- **TrafficType**: Numeric identifier for the type of traffic (e.g. organic search, social media, direct traffic, referrals).
- **VisitorType**: A categorical variable that distinguishes between ‘Returning Visitor’ (regular user), ‘New Visitor’ (new user) and ‘Other’.
- **Weekend**: A Boolean variable that specifies whether the shopping session took place at the weekend.

5. Target – Dependent variable:

- **Revenue**: Boolean target variable (True/False). Indicates whether the session resulted in an actual purchase. It exhibits significant class imbalance.

1.3 Project Objectives

The main objective of this study is to develop and validate a predictive model capable of classifying users’ purchase intent during an e-commerce session. Formally, the problem is formulated as a binary classification task, in which the target variable $Y \in \{0, 1\}$ (represented by the feature **Revenue**) must be estimated from an input vector $\mathbf{x}$ containing 17 heterogeneous features.

A key challenge in this analysis is the significant class imbalance inherent in the dataset, where sessions ending in a purchase (**Revenue** = True) constitute a significant minority compared to sessions involving browsing only. Therefore, the project does not aim solely to maximise overall accuracy, but focuses on optimising more robust performance metrics, which are essential for accurately capturing consumer behaviour in real-world business scenarios.

2 Explorational Data Analysis (EDA)

2.1 Numerical Features Analysis

The dataset contains no null values, and only around 1% of the total samples are duplicates.

The helper function `compute_descriptive_stats` created in the block 1.4 is used to extract the key statistical information regarding the numerical features.

Feature	Mean	Std Dev	Min	Q1	Median	Q3	Max	CV (%)	Skewness
Administrative	2.32	3.32	0.00	0.00	1.00	4.00	27.00	143.48	1.96
Admin. Duration	80.82	176.78	0.00	0.00	7.50	93.26	3398.75	218.74	5.62
Informational	0.50	1.27	0.00	0.00	0.00	0.00	24.00	252.23	4.04
Info. Duration	34.47	140.75	0.00	0.00	0.00	0.00	2549.38	408.30	7.58
ProductRelated	31.73	44.48	0.00	7.00	18.00	38.00	705.00	140.16	4.34
Prod. Duration	1194.75	1913.67	0.00	184.14	598.94	1464.16	63973.52	160.17	7.26
BounceRates	0.02	0.05	0.00	0.00	0.00	0.02	0.20	218.50	2.95
ExitRates	0.04	0.05	0.00	0.01	0.03	0.05	0.20	112.82	2.15
PageValues	5.89	18.57	0.00	0.00	0.00	0.00	361.76	315.29	6.38
SpecialDay	0.06	0.20	0.00	0.00	0.00	0.00	1.00	323.82	3.30

Table 1: Descriptive statistics for the numerical features of the dataset

It is immediately apparent from the table 1 that, for each feature, the mean is higher than the median. We would therefore expect the outliers to have a far from negligible influence, with a small number of users skewing the mean values without representing typical behaviour. This observation is also confirmed by other parameters in the table. The coefficient of variation

$$CV = \frac{\text{Std Dev}}{\text{Mean}} \times 100$$

exceeds the critical threshold of 100% in every instance, suggesting a very high degree of data dispersion. Equally informative is the skewness value, which, remaining almost consistently above 2, indicates severe asymmetry, with a long right-hand tail in each distribution. This analysis therefore immediately provides an overview of the tasks to be carried out during the pre-processing phase:

- the difference in scale between the features requires mandatory scaling,
- the severity of the asymmetry and the frequency of null values suggest the need for a `log1p` transformation (where $\log1p = \log(1 + x)$),
- the extremely high CV values suggest that outliers need to be carefully managed.

The findings of the statistical study are confirmed by an examination of the distribution plots: by using the `plot_numeric_distributions` function in the block 1.5 it is possible to visualize the distribution of numerical features through KDE computed on `log1p`-transformed data, separating the sessions by target class.

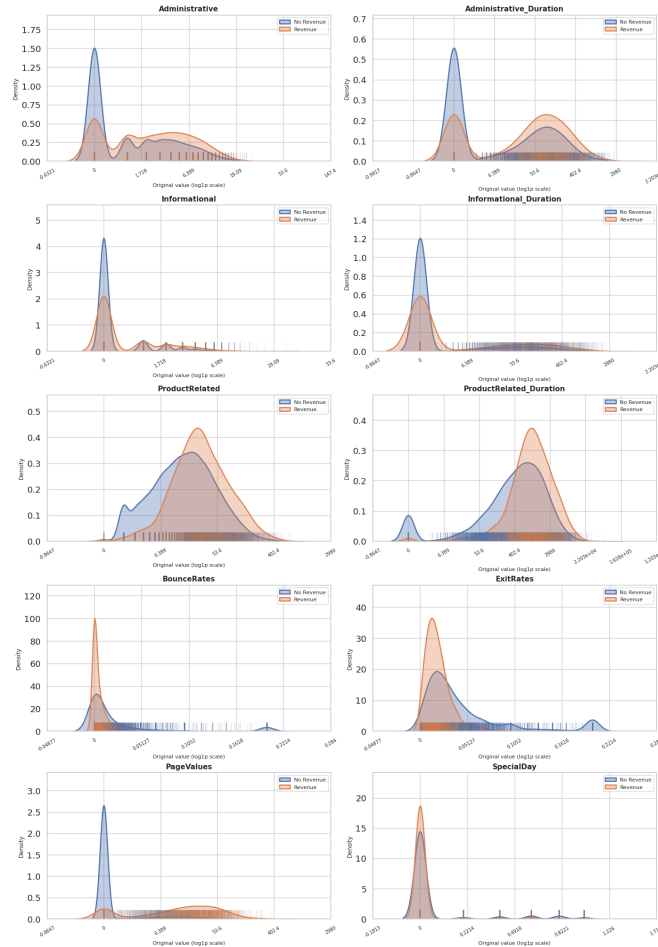


Figure 1: Numerical Feature Distributions by Revenue Class (KDE log1p)

The negative values shown in fig. 1 do not exist in the actual dataset; they are merely a graphical artifact: for reasons of mathematical symmetry, the KDE algorithm displays both the positive portion and its negative mirror image, smoothing out the density of the numerous zero values by distributing them across this central range between -1 and 1 .

As well as demonstrating that right-skewness is an intrinsic characteristic of the features, independent of the outcome, the breakdown by target reveals that the features do not contribute equally to the classification. In particular, `PageValues` shows a clear separation between the probability densities, suggesting significant predictive

potential. Conversely, features such as `SpecialDay` show almost complete overlap, indicating that, taken on its own, this variable has limited predictive power with regard to immediate purchase intention.

2.2 Categorical Features Analysis

For the analysis of categorical features, the conversion rate is introduced as the reference metric:

$$CR = \frac{\text{number of sessions with purchases}}{\text{total number of sessions}}.$$

By evaluating the CR for each attribute alongside the count (using `build_categorical_summary` in block 1.6), it is possible to measure the discriminatory power of each feature in relation to the target.

`Month` (table 2) and `Visitor Type` (table 3) are the features with the most significant signals.

Month	Count	Percentage (%)	Conversion Rate (%)
May	3364	27.28	10.85
Nov	2998	24.31	25.35
Mar	1907	15.47	10.07
Dec	1727	14.01	12.51
Oct	549	4.45	20.95
Sep	448	3.63	19.20
Aug	433	3.51	17.55
Jul	432	3.50	15.28
June	288	2.34	10.07
Feb	184	1.49	1.63

Table 2: Feature analysis: `Month`

Visitor type	Count	Percentage (%)	Conversion Rate (%)
Returning Visitor	10551	85.57	13.93
New Visitor	1694	13.74	24.91
Other	85	0.69	18.82

Table 3: Feature analysis: `VisitorType`

As regards `Month`, there is a marked seasonal pattern. May (27.3%) and November (24.3%) account for over 50% of total traffic, coinciding with two of the most significant commercial events: Mother’s Day and Black Friday. In particular, November also has the highest conversion rate (25.3%), marking the peak of an upward trend that began in October, which ranks second (20.9%). Although it includes Valentine’s Day, February records very low volumes (1.5%) and conversion rates (1.6%).

Among the features, this month may therefore represent one of the strongest predictors. Note the absence of months such as January and April, which suggests that data collection was incomplete during the year.

`VisitorType` presents an interesting paradox. Regular visitors dominate the dataset (85.5%), but NewVisitors have a significantly higher conversion rate (24.9% vs 13.9%). If a user is new, their statistical probability of completing a purchase during that session is almost double. This suggests that regular visitors often use the site for ‘window shopping’ without making a purchase.

Features such as `OperatingSystems`, `Browser` and `Region`, on the other hand, are less informative.

OS ID	Count	Percentage (%)	Conversion Rate (%)
2	6601	53.54	17.50
1	2585	20.97	14.66
3	2555	20.72	10.49
4	478	3.88	17.78
8	79	0.64	21.52
6	19	0.15	10.53
7	7	0.06	14.29
5	6	0.05	16.67

Table 4: Feature analysis: **OperatingSystems**

Browser ID	Count	Percentage (%)	Conversion Rate (%)
2	7961	64.57	15.36
1	2462	19.97	14.83
4	736	5.97	17.66
5	467	3.79	18.42
6	174	1.41	11.49
10	163	1.32	19.63
8	135	1.09	15.56
3	105	0.85	4.76
13	61	0.49	26.23
7	49	0.40	12.24
12	10	0.08	30.00
11	6	0.05	16.67
9	1	0.01	0.00

Table 5: Feature analysis: **Browser**

Region ID	Count	Percentage (%)	Conversion Rate (%)
1	4780	38.77	16.13
3	2403	19.49	14.52
4	1182	9.59	14.81
2	1136	9.21	16.55
6	805	6.53	13.91
7	761	6.17	15.64
9	511	4.14	16.83
8	434	3.52	12.90
5	318	2.58	16.35

Table 6: Feature analysis: **Region**

For **OperatingSystems** (table 4) and **Browser** (table 5) , the dominance is overwhelming, with distributions heavily skewed towards operating **System** 2 (53.5%) and **Browser** 2 (65%). Categories with very few samples show conversion rates that appear extreme, but these are influenced by the low sample size. The stability of the conversion rate across the most popular categories (13–16%) indicates that the operating system and browser are weak predictors compared to other variables. During modelling, the rarest categories could cause overfitting; therefore, it may be advisable to consider grouping or excluding them.

For all values of the variable **Region** (table 6), the conversion rate appears remarkably consistent, fluctuating almost everywhere between 13% and 16%. Such a low variance in the CR between the groups means that the geographical variable also has little discriminatory power.

Traffic ID	Count	Percentage (%)	Conversion Rate (%)
2	3913	31.74	21.65
1	2451	19.88	10.69
3	2052	16.64	8.77
4	1069	8.67	15.43
13	738	5.99	5.83
10	450	3.65	20.00
6	444	3.60	11.94
8	343	2.78	27.70
5	260	2.11	21.54
11	247	2.00	19.03
20	198	1.61	25.25
9	42	0.34	9.52
7	40	0.32	30.00
15	38	0.31	0.00
19	17	0.14	5.88
14	13	0.11	15.38
18	10	0.08	0.00
16	3	0.02	33.33
12	1	0.01	0.00
17	1	0.01	0.00

Table 7: Feature analysis: **TrafficType**

TrafficType (table 7) shows the greatest variation in conversion rate (from 5% to 27%). **TrafficType 2** is very informative, with high volume (32%) and high conversion (22%), whilst **TrafficType 3** presents greater challenges, with high volume (17%) but low conversion (8.8%). The low volume characteristic of the remaining types relegates them to mere browsing of little significance. Overall, this variable can be an important factor in distinguishing high-quality sessions from those of lesser interest.

Weekend	Count	Percentage (%)	Conversion Rate (%)
False	9462	76.74	14.89
True	2868	23.26	17.40

Table 8: Feature analysis: **Weekend**

The table 8 for **Weekend** suggests that around 76% of sessions take place on weekdays. However, the conversion rate increases at the weekend (17.4% compared with 14.9%), suggesting the presence of a slight ‘weekend effect’ on the propensity to purchase.

2.3 Target Analysis and Class Imbalance

To conclude the section on feature description, please refer to the graphs in fig. 2 relating to the analysis of the target feature, **Revenue**.

The following results were obtained by implementing `compute_class_imbalance` and `plot_class_distribution` functions in block 1.7.

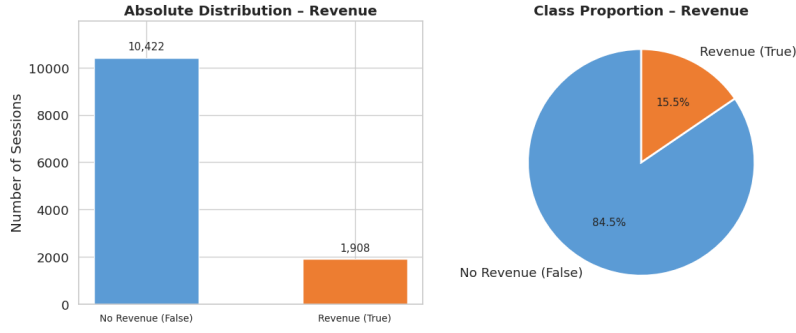


Figure 2: Class Imbalance: target variable 'Revenue'

With a ratio of 5.46:1, this feature shows a significant imbalance between the two classes: 84.5% 'No Revenue' versus 15.5% 'Revenue'. In an e-commerce context, this is normal and expected behaviour: the vast majority of visits to a website do not result in an immediate purchase. From a methodological perspective, this characteristic can have significant implications during the modelling phase, both in terms of potential data augmentation techniques and in terms of variations in the hierarchy of accuracy metrics and interpretations of the confusion matrix.

2.4 Time Context and Special Days

The analysis of the `SpecialDay` variable in block 1.8 reveals a complex data structure that defies initial intuition. Although one might expect that the proximity to an event would boost sales, the data suggest a more nuanced relationship between purchase intention and exploratory browsing.

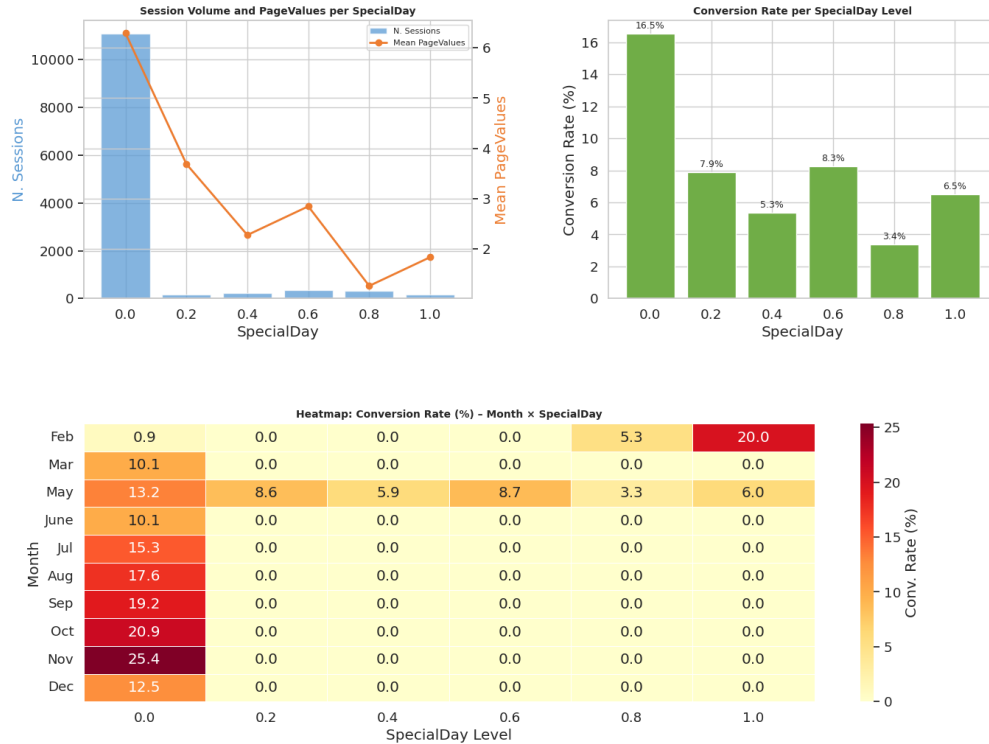


Figure 3: Time Context and Special Days

Looking at the overall data, a downward trend emerges: the Conversion Rate (CR) and the average `PageValues` peak when `SpecialDay` is 0.0 (CR 16.5%, `PageValues` 6.29%), probably due to the dominant volume at that value. This decline as the event approaches (reaching a low of 0.8 with a CR of 3.4%) suggests 'digital window shopping' behaviour. Users visit the site in the days leading up to the holiday to compare products and prices, but without

completing an order. The slight rebound in the CR at **SpecialDay** = 1.0 (6.5%) compared to 0.8 indicates the presence of a group of last-minute buyers, likely driven by the imminent deadline for guaranteed delivery.

The monthly breakdown provided by the heatmap (fig. 3) is crucial: it shows that the impact of **SpecialDay** is not universal, but highly context-dependent. In February, **SpecialDay** is the main predictive driver. On normal days (0.0), the conversion rate is almost zero (0.9%). At the peak of the event (1.0), it shoots up to 20.0%. In May, the pattern is the opposite. Normal days convert better (13.2%) than those influenced by the holiday, a sign of healthy organic traffic being diluted by visitors less inclined to make a purchase during the promotional period. Although it is the month with the highest performance (25.4%), in November the value of **SpecialDay** remains 0. This indicates that events such as Black Friday and Cyber Monday are not mapped to this feature and are treated by the model as standard seasonal traffic. Moreover, the fact that the heatmap shows many cells at 0 for **SpecialDay** greater than 0 means that the **SpecialDay** variable has only been populated for specific months (February, May) and its effect cannot be generalised elsewhere.

A comparison of the first two graphs shows that there is an almost perfect correlation between the decline in **PageValues** and that of the Conversion Rate as **SpecialDay** increases. The sharp drop in the average number of pages viewed before a transaction confirms that pages viewed around public holidays are far less effective in driving transactions than targeted browsing on ordinary days.

2.5 Google Analytics Metrics

To gain a deeper understanding of the data's informational potential, a statistical and visual analysis of Google Analytics metrics may prove useful.

Trough the functions defined in block 1.9 there were obtained the following results.

Metric	Statistics	Revenue: False	Revenue: True	Average Δ (%)
BounceRates	Mean	0,0253	0,0051	-79,80%
	Median	0,0043	0,0000	
	Std	0,0519	0,0122	
ExitRates	Mean	0,0474	0,0196	-58,70%
	Median	0,0286	0,0160	
	Std	0,0512	0,0165	
PageValues	Mean	1,9760	27,2645	+1279,80%
	Median	0,0000	16,7581	
	Std	9,0724	35,1920	

Table 9: Descriptive statistics for Google Analytics metrics grouped by Revenue

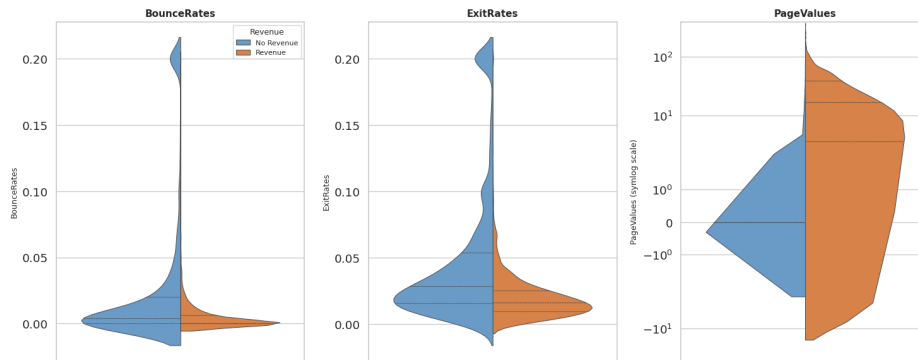


Figure 4: GA Metrics - Violin Plot in log-scale split by Revenue Class

The presence of negative values in the distributions shown refers to the same phenomenon described for fig. 1.

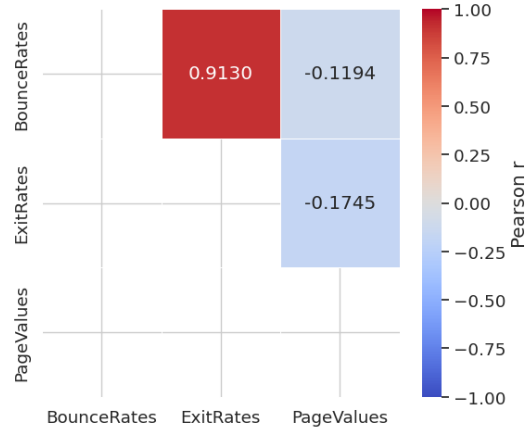


Figure 5: GA Metrics – Pearson Correlation

The **PageValues** metric emerges as the most significant predictor across the entire dataset. As shown in table 9, there is an increase in the mean of +1279.8% for the **Revenue = True** class (27.26) compared with the False class (1.98). The distribution shown in the Violin Plot (fig. 4) confirms that, whilst almost all users who do not make a purchase have a **PageValues** of zero, the positive class is distributed across significantly higher values. Conversely, sessions that generate revenue show drastically lower drop-off rates: **BounceRates** plummet by 79.8% and **ExitRates** by 58.7%. This suggests that purchase intent is correlated with in-depth and targeted browsing. A user intending to make a purchase does not leave the site on the first page (low **Bounce**) and tends to navigate through several steps before leaving (low **Exit**).

Significant critical evidence is also provided by the correlation heatmap (fig. 5), which reveals a Pearson Correlation coefficient of $r = 0.91$ between **BounceRates** and **ExitRates**. This level of collinearity indicates redundancy in the data, which may warrant consideration of data pre-processing prior to the modelling phase.

2.6 Multivariate Correlation

To gain a definitive understanding of the linear relationships between variables, analysing the Pearson Correlation matrix is a crucial step.

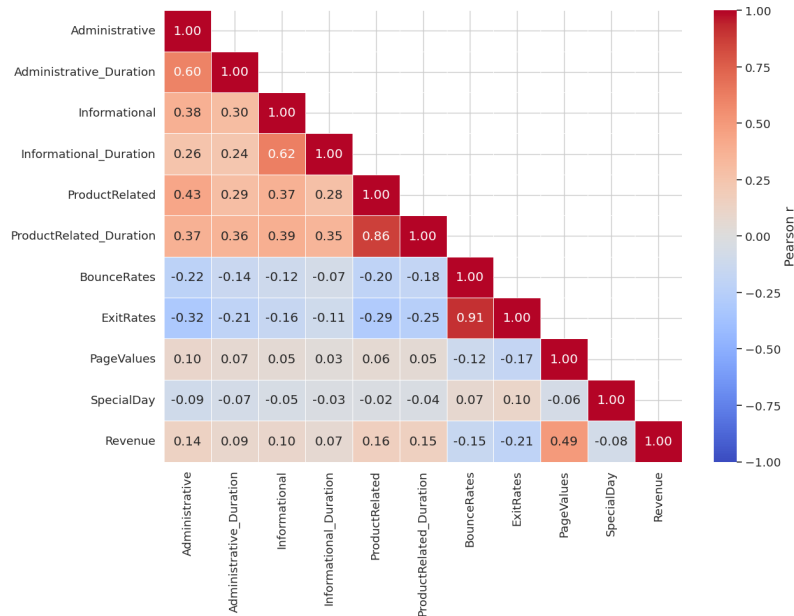


Figure 6: Correlation Matrix

An analysis of the heatmap obtained by the `compute_global_correlation` helper (block 1.10) reveals three main clusters of relationships.

1. Correlation between Activity and Duration

There is a significant positive correlation between the number of pages visited and the time spent for each category: **ProductRelated** (0.86), **Administrative** (0.60) and **Informational** (0.62) with their respective durations. Clearly, if a user increases the number of pages visited, the total time increases almost linearly. These pairs of variables may provide redundant information.

2. Multicollinearity Google Analytics metrics

The correlation between **BounceRates** and **ExitRates** is the most statistically significant ($r = 0.91$). This suggests that the two features describe almost the same phenomenon of ‘abandonment’.

3. Relationship with the Target variable

Looking at the bottom row of the matrix, we can rank the linear influence of the features on conversion. **PageValues** is confirmed as the only strong linear predictor ($r = 0.49$). **ExitRates** (-0.21) and **BounceRates** (-0.15) show a negative correlation, confirming that as exit rates increase, the probability of purchase decreases. Variables such as **SpecialDay** (-0.08) and the durations of information/administrative sessions have correlations close to zero.

It is important to emphasise that a low linear correlation does not necessarily indicate that the variables are of little use: they may underlie a non-linear relationship with the target variable, which only more complex models will be able to capture.

3 Pre-processing and Dimensionality Reduction

The pre-processing phase is a necessary step in transforming the raw dataset into a representation space suitable for training algorithms. Due to the inherently heterogeneous nature of the variables, it is essential to apply a sequential pipeline of data engineering and data cleaning.

The following subsections will describe the mathematical and algorithmic operations applied.

3.1 Categorical Encoding and Feature Selection

To make the dataset interpretable by supervised learning algorithms, in block 2.1 a One-Hot Encoding (OHE) strategy was adopted for nominal variables and a binary conversion for Boolean variables. The choice of OHE is motivated by the non-ordinal nature of the features: variables such as **OperatingSystems** or **Region** do not have an intrinsic order, and the assignment of sequential numerical labels (1, 2, 3...) could lead the model to mistakenly perceive non-existent order relationships or metric proximity, thereby adversely affecting performance. The encoding process conducted by the `encode_features` helper has resulted in a significant transformation of the dataset’s structure.

The dataset has expanded from the original 18 columns to 75 final features, with an increase of 57 columns due to the high cardinality of variables such as **TrafficType** (20 categories) and **Browser** (13 categories).

Many of these columns, however, belong to extremely rare categories (as can be seen in table 3, table 4, table 5). The decision was to apply a feature selection strategy by setting a variance threshold of 0.01. A threshold of 0.01 means that all features with a volume of less than 1% are removed.

The aim is twofold: to prevent overfitting and to ensure computational efficiency. The handling of continuous numerical variables was crucial: as these are non-binary variables, their variance is on a different scale and should not be filtered using the same criteria as dummy variables.

Through the `remove_low_variance` function in block 2.2, 20 uninformative binary features, attributable to residual or highly minority categories, were excluded from the dataset: specifically, secondary browsers (3, 7, 9, 11, 12, 13), niche operating systems (5, 6, 7, 8), marginal traffic sources (such as channels 7, 9 and 12, and the range from 14 to 19), and the residual user class **VisitorType_Other** were removed. The model will therefore focus on the remaining classes of the features affected by this reduction.

3.2 Feature Scaling

The standardisation of continuous variables is an essential step, particularly when working with data on different scales and implementing models based on the calculation of euclidean distances or on variance maximisation, such as dimension reduction techniques (PCA) or clustering algorithms (K-Means).

As the *Online Shoppers Purchasing Intention* dataset is characterised by intrinsically heterogeneous feature distributions and units of measurement, the **StandardScaler** was applied in block 2.3.

This operator performs an affine transformation based on the calculation of the Z-score

$$z = \frac{x - \mu}{\sigma},$$

centring the data on a mean of zero ($\mu = 0$) and scaling it to a standard deviation of one ($\sigma = 1$).

In order to validate the effectiveness and impact of the transformation, by `plot_scaling_comparison` in block 2.4 a visual comparison was produced between the original distributions and the scaled distributions for the ten continuous numerical features in the dataset. Whilst the pre-scaling representations show the data in their original absolute metrics (seconds, page counts, percentages), the post-scaling graphs reproject the same information by measuring it in terms of squared deviations from the global mean.

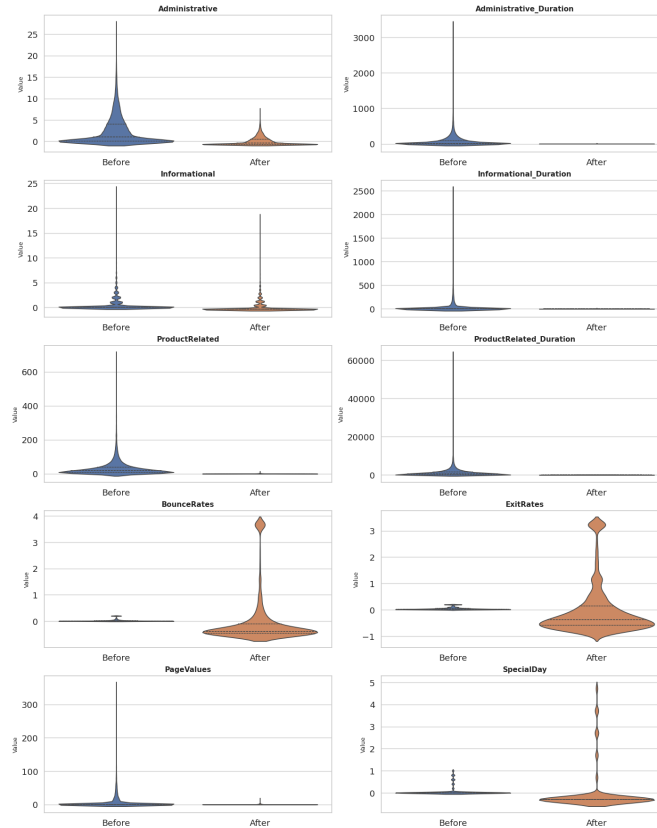


Figure 7: Before and After Comparison following the application of StandardScaler.

A visual analysis of the violin plots in fig. 7 allows the theoretical properties of standardisation to be verified empirically.

If we look at the y-axes in the various graphs, it is clear that there is a complete lack of consistency in scale between the different features in their pre-scaling distribution. Turning our attention to the post-scaling graphs, we can see that all variables are now confined to a consistent and comparable range of variation. This balance is essential to prevent variables with intrinsically larger orders of magnitude from dominating the updating of gradients or the calculation of covariance matrices.

In terms of asymmetry and density concentration, a comparison of the shapes of the violins between the previous

and subsequent phases reveals that they overlap perfectly. This confirms that the Z-score preserves the original distributional information and the geometry of the data intact: the transformation changes the unit of measurement, but does not distort the statistical nature of the variable in any way.

The presence of very long ‘necks’ that taper off towards the higher values (particularly evident in high-variance features such as *PageValues* or the *Duration* metrics) indicates the presence of heavy tails composed of anomalous sessions. It is essential to understand that the **StandardScaler** neither filters nor attenuates such anomalies: they are faithfully preserved, but repositioned within a standardised metric where their distance from the data’s centre of gravity is mathematically weighted relative to the behaviour of the entire population.

3.3 Principal Component Analysis (PCA)

Following the encoding and filtering operations, the dataset consists of 55 features, with an inevitably strong intrinsic correlation that risks distorting the effectiveness of algorithms based on metric distances.

PCA is introduced in block 2.5 to mitigate this issue, with the dual objective of reducing dimensionality whilst preserving a significant proportion of the original information, and of projecting the data into a new latent space of orthogonal and uncorrelated variables, ensuring that each component captures a unique and non-redundant information signal.

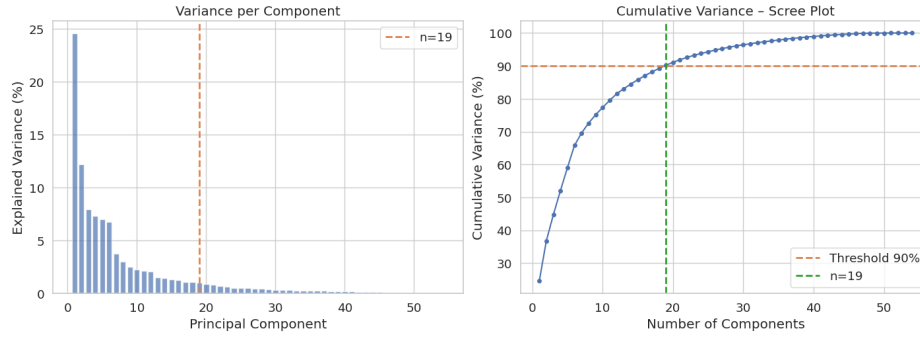


Figure 8: PCA – Analysis of explained variance

By setting a threshold to retain 90 per cent of the total variance, the algorithm selected 19 principal components, resulting in a reduction in the original dimensionality of approximately 65% – that is, from 55 to 19 dimensions in total.

The first principal component (PC1) alone accounts for almost 25% of the total variance, which highlights the presence of a primary and dominant axis in user behaviour (likely attributable to the intensity of interaction with the platform). On the other hand, the need to extend the selection to 19 components in order to account for 90% of the variance demonstrates the inherently multidimensional nature of the browsing experience and purchase intent. Therefore, there is no single dominant factor, but rather a dispersion of the predictive signal across a variety of behavioural, temporal and technical dimensions.

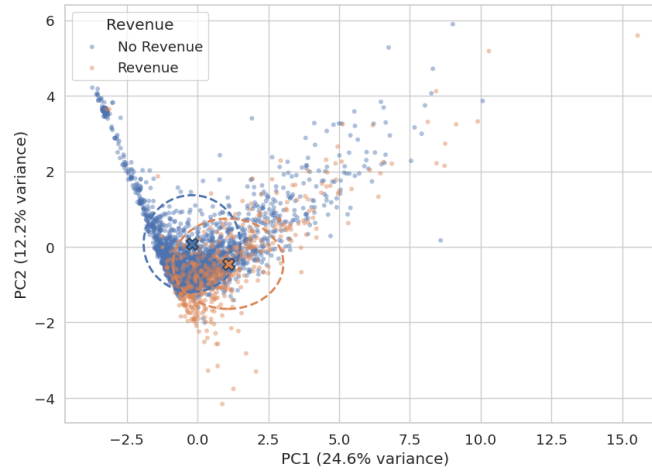


Figure 9: PCA – Class separability (PC1 × PC2)

From the perspective of the geometric separability of the classes, analysis of the two-dimensional plane defined by the first two principal components in fig. 9 reveals a partial but clear spatial differentiation between their respective centroids. The centroid associated with the successfully completed sessions is significantly shifted towards positive values of PC1 compared with the group where **Revenue = False**. Nevertheless, there is a large area of overlap between the confidence ellipses of the two classes, indicating that when the data are projected onto just the first two dimensions, many unsuccessful sessions retain a geometric profile that is statistically indistinguishable from those that generate revenue.

The overall distribution of points takes on a distinctive fan-shaped or triangular pattern. The dense concentration of negative observations in the top-left quadrant identifies an extremely specific user profile, associated with visits characterised by a high bounce rate or minimal time spent on the site, which almost never result in a transaction. Given the spatial location of the centre of gravity of conversions, it can be inferred that PC1 is strongly governed by variables such as Page Value and the time spent on product pages (**Product Related Duration**); consequently, a user's progression towards the far right along the PC1 axis corresponds to a systematic and robust increase in their probability of conversion.

The superimposition of the ellipses also highlights that a simple linear model would exhibit marked limitations in performance, generating a high number of false positives. To maximise predictive accuracy, it will therefore be essential to utilise the full range of the 19 previously selected components or, alternatively, to rely on non-linear ensemble algorithms such as Random Forest or XGBoost.

Finally, the outliers extending towards the far right ($PC1 > 10$) and upwards ($PC2 > 4$) reflect exceptional purchasing sessions, characterised by hundreds of views or anomalous dwell times; at these extremes, the empirical density of the conversion class increases proportionally more.

Loadings Interpretation

The analysis of projection coefficients, or loadings, included in block 2.6 forms the link between the mathematical decomposition into eigenvectors and the semantic interpretation of purchasing behaviour on the portal. By examining the influence of the original variables on the formation of the principal components, it is possible to relate the abstract space generated by PCA to concrete and interpretable behavioural dimensions.

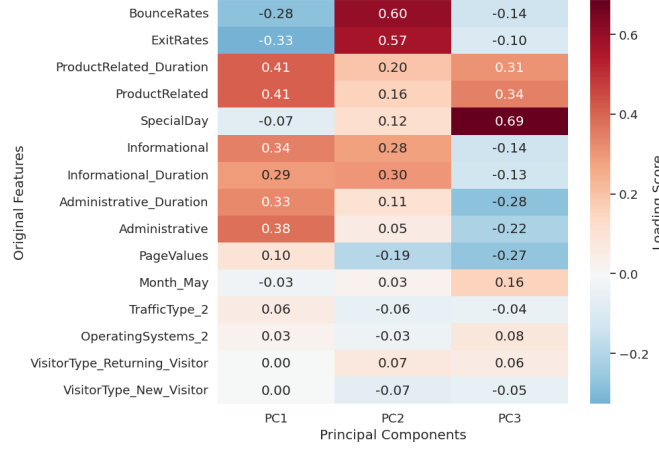


Figure 10: PC Interpretation: Top 15 influential features

The first principal component (PC1) exhibits high and balanced positive coefficients for the variables **Product Related Duration** (0.41), **Product Related** (0.41), **Administrative** (0.38) and **Informational** (0.34). At the same time, it exhibits significant negative weights for **Exit Rates** (-0.33) and **Bounce Rates** (-0.28). Consequently, a session characterised by a high score along PC1 describes a user engaged in proactive and extensive exploration of the platform, with a prolonged visit duration and a marked curiosity for information. The strong inverse correlation with the drop-off metrics allows PC1 to be characterised as the ‘Engagement Volume’ dimension, confirming it as the principal axis of data variability.

The second principal component (PC2), on the other hand, is dominated almost exclusively by **Bounce Rates** (0.60) and **Exit Rates** (0.57). High scores on this axis unequivocally identify transient sessions or those lacking genuine interest, in which the user leaves the platform almost instantly. It is interesting to note that the variables relating to information pages have a positive coefficient on this component, which may reflect visits aimed solely at a quick, disengaged browse that does not feed into the conversion funnel. PC2 can therefore be interpreted semantically as the session’s ‘Negative Quality’ index.

With regard to the third principal component (PC3), a polarisation of the entire matrix is observed in relation to the **Special Day** variable (0.69). PC3 isolates the knock-on effect exerted by public holidays or special occasions in an extremely targeted manner. A high value for this component indicates that the purchasing session is strongly influenced by its proximity to an event. However, the fact that the **Page Values** variable (-0.27) shows a negative loading on this coordinate suggests caution: the traffic generated during **Special Days**, whilst quantitatively significant, may reflect browsing aimed at finding gift ideas or comparing prices, without necessarily resulting in an immediate high-value conversion.

In summary, PCA allows the volume and time dimensions (PC1) to be synthesised in a coherent manner and clearly separated from the drop-off metrics (PC2), confirming the high multicollinearity that characterised the original data space. The latent model clearly distinguishes between browsing intensity (PC1), the speed at which users exit the site (PC2) and dependence on the festive context (PC3). Finally, characteristics relating to visitor type or the operating system used show loadings close to zero on all the first three components. This demonstrates that these variables contribute only marginally to the dominant variance in the dataset, being explained solely by subsequent latent axes and confirming their nature as secondary information or noise relative to direct web interaction metrics.

3.4 t-SNE Non-linear Exploration

To explore the underlying structure of users’ browsing behaviour and identify potential non-linear patterns that precede conversion, it is necessary to complement traditional linear metrics with a manifold learning technique. To this end, the t-SNE (*t-Distributed Stochastic Neighbour Embedding*) algorithm was applied; this is a tool for projecting complex, high-dimensional datasets into a two-dimensional space whilst preserving their local neighbourhood relationships.

Theoretically, the t-SNE algorithm converts distances between observations into similarity measures, ensuring that nearby points in the original data remain close in the new two-dimensional projection. In mapping this information,

the algorithm optimizes the arrangement of the points to preserve local relationships and avoid the crowding problem at the center of the map. This mathematical approach ensures a balanced visual representation, allowing data clusters to emerge more clearly.

As this is an algorithm based on distance calculations, the standardisation of continuous numerical features is a fundamental prerequisite to ensure that variables with different magnitudes do not unduly dominate the spatial metric. Furthermore, given the computational complexity of t-SNE ($\mathcal{O}(N^2)$), the algorithm was applied in block 2.7 following a linear reduction via PCA and on a stratified sample of the dataset, ensuring that the original proportions of the target **Revenue** were preserved.

A key hyperparameter for the correct tuning of t-SNE is perplexity, which can be interpreted as a smoothed measure of the effective number of local neighbours for each point. Low values cause the algorithm to focus on the strictly local structure (generating many small clusters), whilst high values force the algorithm to take account of the global structure of the dataset. In order to avoid visual artefacts and misinterpretations, the analysis was conducted by assessing the stability of the projection as this parameter varied: the implemented `compute_tsne_grid` allowed to compute the t-SNE projections for each perplexity value in a list, as following.

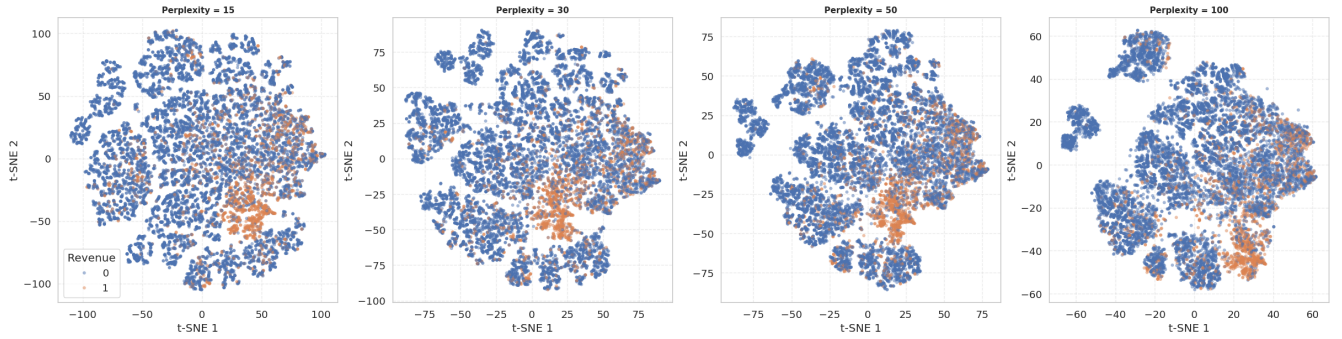


Figure 11: t-SNE – Perplexity Comparison in PCA Space

A visual analysis of the resulting projection, shown in fig. 11, provides valuable insights into the dynamics of the conversion funnel. By examining the grid as the perplexity varies, it is possible to draw a number of methodological and business-related conclusions.

If we look at the graphs with low perplexity (e.g. 15 and 30), we can see a highly fragmented structure, consisting of discrete micro-islands. This indicates extreme variability in local navigation behaviour. As perplexity increases, the micro-islands converge into more cohesive macro-structures. The persistence of certain geometric shapes at high perplexity levels (such as the lower ‘peninsula’) confirms that this clustering is not a statistical artefact, but a real and robust pattern in the data.

Regardless of the value selected, a cluster of orange points – which are almost isolated – consistently emerges; these identify sessions with positive **Revenue**, tending to be located in the lower or lower-right quadrant.

If such a stable positioning is observed in a space generated solely by continuous features, it means that for this specific subgroup of users, the quantitative signals (likely very high *Page Values* or specific dwell times on product pages) are strong enough to define purchase intent in an almost deterministic manner. This cluster represents the ‘typical’ or highly motivated buyer.

Turning to the central and upper macro-cloud, the scattering of orange points is overshadowed by the massive presence of blue points (**Revenue = False**). The direct implication of this spatial coexistence is that browsing dynamics alone (bounce times and frequencies) are not sufficient to distinguish the outcome of the session. For these users, the browsing behaviour of those who make a purchase is identical to that of those who abandon the site; therefore, the key to correctly classifying these sessions necessarily lies in the categorical variables excluded from this projection (such as **Special Day**, **Month**, **VisitorType** or **TrafficType**).

In conclusion, the t-SNE analysis demonstrates the need to use complex machine learning models: a simple linear model based on continuous traffic metrics would only be able to capture the isolated ‘pure’ cluster, inevitably failing to disentangle the complex overlap present in the rest of the vector space.

4 Clustering and Feature Engineering

The application of unsupervised learning techniques, prior to the classification stage itself, is an essential analytical strategy for exploring latent structures within web traffic.

The K-Means algorithm was used to partition the user space into homogeneous clusters by minimising inertia, i.e. intra-cluster variance. As it is based on the calculation of euclidean distances, K-Means is extremely sensitive to dimensionality. For this reason, the clustering was trained on the principal components extracted previously, ensuring an orthogonal metric space, free from multicollinearity and cleansed of background noise.

The choice of the *K*-Means algorithm over *K*-Medoids is motivated by the former’s superior computational efficiency when applied to the dataset’s more than 12,000 instances and by the geometric nature of the input space. By operating on continuous coordinates that have been scaled and reduced via PCA, the definition of the centroid based on the euclidean mean ensures the stability and interpretability of representative average behavioural profiles, which are preferable to anchoring to individual real samples from the dataset.

The aim of this phase is not to predict the target variable directly, but rather to extract a new behavioural macro-feature that summarises the user’s browsing behaviour, thereby enriching the subsequent supervised models.

4.1 Optimal Cluster Number (*k*)

Determining the optimal number of centroids *k* requires striking a balance between maximising statistical cohesion and ensuring genuine behavioural interpretability from an e-commerce perspective. To this end, two internal validation metrics were analysed: the Elbow Method (based on inertia) and the calculation of the Silhouette Score. By `evaluate_kmeans_internal` in block 2.8 it was possible to compute both inertia and Silhouette for each *k* in a certain range.

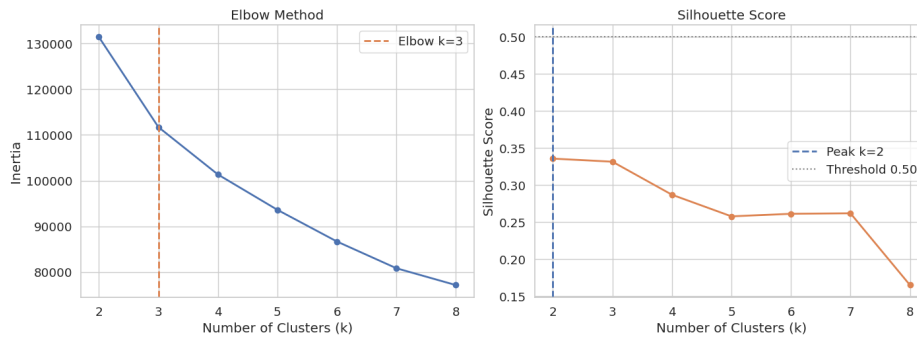


Figure 12: K-Means – Selecting the optimal *k*

Looking at the inertia curve, a marked and continuous decline can be observed. Although no clear ‘elbow’ is apparent, there are dips in the rate of decline at $k = 3$ and $k = 4$, suggesting that beyond these thresholds, the addition of further centroids fragments the data without providing any substantial informational gain. The Silhouette Score plot, on the other hand, provides a rigorous indication of spatial separation: the absolute maximum value is recorded for $k = 2$ (approximately 0.34), after which it declines naturally. The maximisation at $k = 2$ means that, from a purely geometric point of view, the most evident structure in the data is a dichotomy. However, by limiting the analysis to the mathematical indicator alone and ignoring the complexity of the application domain, one might overlook micro-patterns that are essential to the business. For this reason, the investigation was extended in block 2.9 to the values $k = 2$, $k = 3$ and $k = 4$, comparing the partitions obtained in PCA space with the actual empirical distribution of the target variable **Revenue**.

4.2 Clustering Comparison ($k=2, 3, 4$)

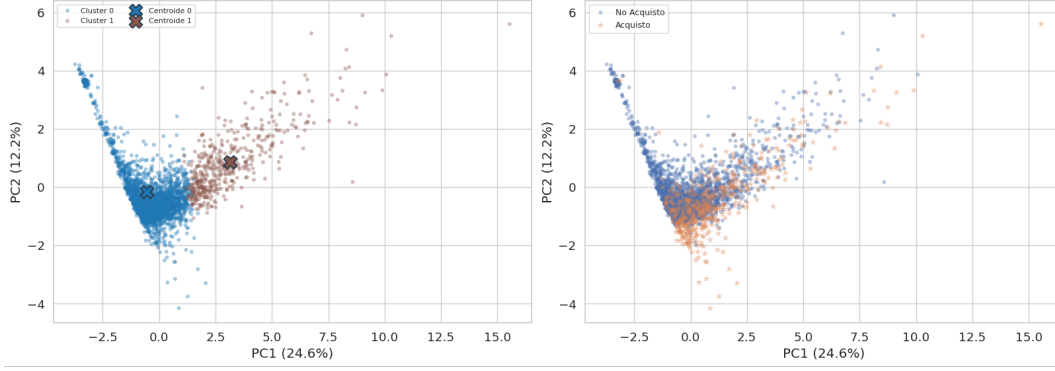


Figure 13: K-Means in PCA space vs **Revenue** distribution ($k = 2$)

By setting the algorithm to search for just two clusters (fig. 13), the two-dimensional space defined by the first two principal components is split vertically. Cluster 0 (in light blue) encompasses the vast majority of standard sessions clustered near the origin, whilst Cluster 1 (in brown) isolates the long tail of observations extending towards the positive values of PC1.

When comparing this partition with the actual distribution of **Revenue**, a critical misalignment becomes apparent. Actual purchases (marked with orange stars) are concentrated in a dense cluster in the lower-central portion of the space, straddling the two clusters proposed by K-Means. Such a weak overlap between the assigned cluster and the positive target class means that the algorithm is simply separating sessions of ‘normal’ duration from extremely long sessions (the exploratory outliers). Therefore, $k = 2$ is insufficient to isolate the true purchase intention, merely mapping the volume of traffic.

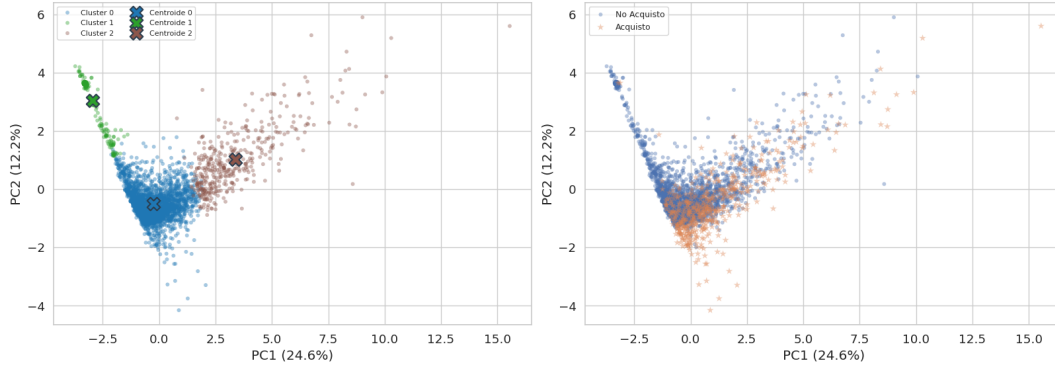


Figure 14: K-Means in PCA space vs **Revenue** distribution ($k=3$)

By increasing the granularity to three centroids (fig. 14), the geometry of the space begins to reveal more sophisticated behavioural patterns. K-Means preserves the separation of the very high-intensity navigators (Cluster 2, in brown), but splits the main dense cluster into two distinct entities: Cluster 0 (in light blue) at the centre and a branch extending upwards to the left, identified as Cluster 1 (in green).

The isolation of this upper-left arm is an extremely valuable analytical signal. Looking at the **Revenue** map, conversions are statistically non-existent in that exact region of the space. The perfect overlap between Cluster 1 and the area with a zero purchase rate suggests that K-Means has, in an unsupervised manner, identified the ‘bouncers’ segment (users with **Bounce Rates** and **Exit Rates** close to 1, or very short sessions). This result demonstrates how clustering can act as a powerful semantic filter, grouping together sessions with no commercial potential.

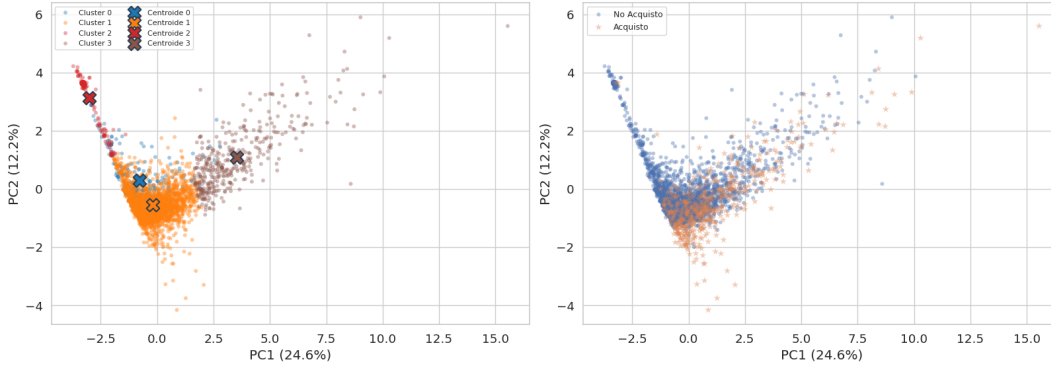


Figure 15: K-Means in PCA space vs **Revenue** distribution ($k=4$)

The four-cluster configuration (fig. 15) represents the optimal point of convergence between mathematical abstraction and predictive utility. In this scenario, the algorithm preserves the identification of the extreme profiles (Cluster 3, brown) and the bouncers (Cluster 2, red), while introducing a fundamental split within the central cloud of organic traffic. This gives rise to a Cluster 0 (light blue, upper portion) and a Cluster 1 (orange, lower portion).

The relevance of this partition becomes evident when compared with the dispersion of the target. The Cluster 1 mapped by K-Means retraces with surprising precision the morphology and density of the sessions that conclude with a transaction. The fact that an unsupervised algorithm is able to geometrically circumscribe the high-conversion zone means that the latent signals captured by the navigation metrics are sufficiently coherent to define a unique behavioral signature. Ultimately, the $k = 4$ partition accurately describes the four souls of e-commerce: immediate exit (bounce), recreational shopping without purchase (window shopping), concrete purchase intent, and anomalous intensive navigation.

The identifier of this cluster will therefore become a categorical feature of high value for the training of the subsequent classifiers.

4.3 Cluster Profiling

The ultimate objective of unsupervised partitioning is not exhausted in mere geometric visualization, but culminates in the creation of a new categorical variable, denominated **Cluster_ID**. This feature acts as a behavioral meta-variable capable of synthesizing the complex interaction among the user's navigation metrics. Analyzing the descriptive statistics of the centroids as k varies allows the translation of mathematical distances into genuine business profiles, evaluating which configuration provides the sharpest predictive signal to be injected into the subsequent classification algorithms.

Setting the algorithm to two partitions, K-Means performs the most basic of fragmentations, dividing the user base according to the absolute intensity of the navigation. The Cluster 0 absorbs the vast majority of the traffic (over 10,400 observations), delineating a "light" user who visits on average approximately 21 product pages and converts at a rate of 13.3%. The Cluster 1, by contrast, isolates a niche of "heavy" navigators (approximately 1,800 users) characterized by extreme volumes: over 91 product pages visited, prolonged dwell times, and a **PageValues** that is nearly double (9.8). If you observe a clear doubling of the conversion rate (27.8%) corresponding to a drastic increase in the volume of pages visited, it means that the model is capturing exclusively the linear correlation between time spent and the probability of purchase. The implication is that the **Cluster_ID** variable thus generated would prove too simplistic for an advanced classifier, since it flattens all the intermediate nuances into a trivial "low vs high traffic" logic.

The introduction of a third centroid radically modifies the topology of the space, providing a crucial informational benefit. While the standard (Cluster 0) and intensive (Cluster 2) traffic profiles retain physiognomies similar to the previous case, the new Cluster 1 intercepts an extremely specific and pathological segment. This group exhibits an almost null conversion rate (0.57%), coupled with anomalous spikes in **BounceRates** (16.6%) and **ExitRates** (18.0%), against merely 3 visited product pages. If one notes the convergence between a **PageValues** equal to zero and maximized exit rates, it means that the algorithm has geometrically isolated the immediate abandonment phenomenon. This significantly enriches the **Cluster_ID** feature: providing a decision tree with a label that identifies

”junk” or disinterested traffic a priori allows the false positives to be drastically reduced during the classification phase.

The four-cluster configuration represents the apex of semantic exploration, revealing dynamics not purely linked to traffic volume, but to the temporal context. Alongside standard traffic (Cluster 1), intensive buyers (Cluster 3, with peaks in the month of November) and bouncers (Cluster 2), an unprecedented Cluster 0 emerges. This segment, composed of almost a thousand users, presents a low conversion rate (6.32%) but is overwhelmingly dominated by two variables: 94.5% of visits occur in May and the average value of `SpecialDay` skyrockets to 0.68. If one observes an almost total concentration of the cluster in proximity to a single recurrence (Mother’s Day), combined with a high number of visited pages but scarce conversions, the implication is that the algorithm has mathematically isolated the phenomenon of festive ”window shopping”. These users browse intensively to look for gift ideas or compare prices, but do not finalize the purchase. The creation of a four-level `Cluster_ID` therefore equips the supervised models with an exceptional magnifying glass, capable of distinguishing those who browse to buy from those who browse solely due to the inertia of a holiday.

5 Data Preparation

5.1 Stratified Train-Test Split

The predictive modeling phase requires a rigorous separation between the data used for training and those intended for performance validation, in order to ensure an unbiased estimate of the models’ generalization capability and to avoid overfitting phenomena. To this end, the overall dataset composed of 12,330 sessions was split into a Train Set and a Test Set according to a classic 80/20 proportion.

From a methodological standpoint, the operation was performed by applying a stratified partition with respect to the target variable `Revenue`. Stratification is a fundamental requirement when operating with heavily imbalanced datasets, since it ensures that the original proportion of the classes is faithfully replicated in both subsets, averting systematic distortions during the learning and testing phases. Furthermore, the split was applied downstream of the clustering process: in this way, the engineered feature `Cluster_ID` is inherited coherently and cleanly in both the train and test sets, without the risk of data leakage.

By using the function `split_scaled` in block 3.1 it was possible to perform a stratified train-test split on the already scaled dataframe, obtaining the following separated sets in table 10.

Dataset	Sample	Features	Class 0	Class 1
<code>X_train_sc</code>	9864	55	84.53%	15.47%
<code>X_test_sc</code>	2466	55	84.51%	15.49%

Table 10: Split Results

The correct execution of the split procedure finds empirical confirmation in the dimensional and distributional analysis of the two generated subsets. The final Train Set is composed of 9,864 sessions, while the Test Set gathers 2,466 of them, maintaining for both a coherent set of 55 features (inclusive of the numerical variables, the categorical dummies and the cluster identifier).

The effectiveness of the stratified sampling is mathematically validated by the perfect alignment of the target variable distribution. In the Train Set, the negative class (`Revenue = 0`) stands at 84.53%, while the positive class (`Revenue = 1`) represents 15.47% of the observations. In the Test Set, a nearly identical distribution is observed, with 84.51% for the negative class and 15.49% for the positive class. This confirms that the model will face during the testing phase the same probabilistic scenario and the same informational asymmetry experienced during the training phase.

Finally, on the level of code architecture, the pipeline was structured to return in parallel both the scaled matrices (`X_train_sc`, `X_test_sc`) and the original matrices (`X_train`, `X_test`). This ”dual-track” architecture preserves the logical separation of the data while ensuring, at the same time, the correct training of the estimators sensitive to feature scale and the immediate physical interpretability of the original variables during the profiling and analysis phases.

5.2 Mutual Information and Feature Selection

Before proceeding with the training of the predictive models, a quantitative evaluation of the informative power of the individual variables with respect to the target **Revenue** becomes necessary. To this end, the *Mutual Information* (MI) was computed on the training set. This decision is dictated by the strongly hybrid nature of the dataset, composed of continuous, discrete and categorical variables, and by the presence of marked non-linear patterns, such as the trend of **SpecialDay** or of the Google Analytics metrics.

The Mutual Information is a non-parametric metric based on Shannon entropy that measures the amount of information shared between two random variables X and Y :

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right)$$

A score equal to zero indicates that the two variables are statistically independent, whereas progressively higher

values indicate a greater capacity to reduce the uncertainty on the target by knowing the value of the feature.

By computing the entropy, the MI provides a standardized and directly comparable metric that allows placing on the exact same level the impact on the target of continuous variables, such as **PageValues**, and categorical ones, such as **VisitorType**. Unlike Pearson's linear correlation coefficient, the MI does not assume linear relationships and is able to capture non-linear dependencies and complex interactions.

To ensure this homogeneous comparison, the algorithm implemented in **Scikit-Learn** included in the `analyze_mutual_information` function in block 3.2 adopts two distinct estimation strategies depending on the nature of the variable, which is distinguished by using a boolean mask.

For discrete variables and categorical dummies, the computation relies on relative frequencies and direct counts to estimate the marginal and joint probabilities. Conversely, for continuous variables the *k-Nearest-Neighbours* (*k*-NN) is employed, a non-parametric algorithm based on the *k* nearest neighbours, which computes the distance between points in the continuous space and thus estimate the local probability density without assuming any prior distribution.

From a methodological standpoint, the MI offers an important complementary value with respect to other dimensionality reduction techniques. While PCA projects the data into a latent space of orthogonal components that are not directly interpretable from a business perspective, the Mutual Information allows preserving the original feature space, drawing up a transparent ranking of the most important variables for conversion.

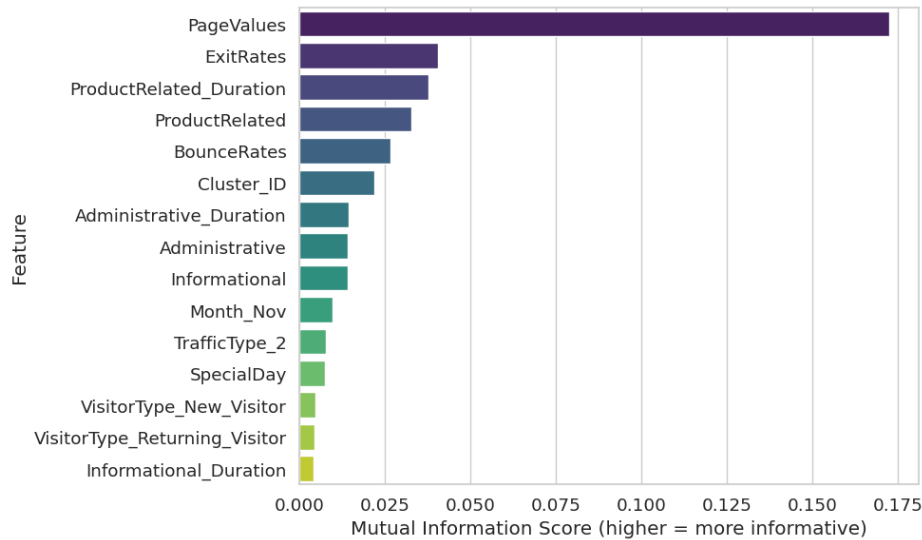


Figure 16: MI scores for the top 15 most informative features with respect to **Revenue**.

The analysis of the empirical results shown in Figure 16 highlights a structural polarization of the informative set, overwhelmingly dominated by the **PageValues** variable. With a score of 0.172, this feature clearly stands out from

any other variable in the dataset, reflecting its nature linked to the economic value attribution performed by Google Analytics.

Below this, a cohesive block of variables linked to active engagement and navigation friction emerges. In this scenario, the propensity to abandon represented by `ExitRates` (0.041) and `BounceRates` (0.027) contrasts with the dwell time on product pages, expressed by `ProductRelated.Duration` (0.038) and the `ProductRelated` count (0.033). It is interesting to note how the overall duration surpasses the number of visited pages in informative power. This suggests that the time spent processing the visual and textual information about the product is a more robust signal of purchase intention compared to a rapid and potentially scattered navigation.

In an intermediate range of relevance, yet still fundamental for the stability of the predictive signal, lie the service or administrative activities, such as `Administrative.Duration` and `Administrative`, alongside `Informational`. These variables, although settling on Mutual Information values close to 0.014, describe accessory actions that are nonetheless typical of pre-transactional phases, such as cart consultation or the reading of shipping conditions.

Finally, the analysis offers a finding of considerable methodological value by examining the positioning of the temporal factors and the engineered variables. The seasonal dummy `Month_Nov` (0.010) and the `Cluster_ID` identifier (0.022) firmly place themselves in the top 10 of the most informative features. While the November score mathematically validates the impact of the major autumn promotional events on purchasing behavior, the value associated with the cluster demonstrates the effectiveness of the process carried out previously. Although it never had access to the target variable during the k -Means fitting phase, the geometric partitioning obtained in the PCA space was able to synthesize a behavioral signal that possesses a marked informative relevance for the purposes of the final classification.

It must nonetheless be highlighted that this approach has an intrinsic limitation: MI evaluates the relationship between each single feature and the target in a purely individual manner, without analyzing how the variables interact with one another or whether there are multivariate informational redundancies.

To this end, starting from the computation of the MI, a two-stage hybrid selection strategy was implemented, aimed at maximizing the informative relevance and minimizing the redundancy.

In the first stage, a relevance filter based on the MI with an extremely conservative minimum threshold $\tau_{MI} = 0.001$ allowed for the exclusion of 31 variables deemed statistically non-informative or containing noise uncorrelated with the target `Revenue`.

In the second stage, for the remaining features, the linear collinearity was evaluated by means of the Pearson correlation matrix, setting a tolerance threshold $\tau_r = 0.80$. In the cases where strong redundancy was found, the Mutual Information was employed as a decision criterion to preserve exclusively the variable with the highest predictive power towards the target class, discarding the less informative counterpart.

This process led to the elimination of 3 redundant features:

- **ProductRelated**: excluded in favor of `ProductRelated.Duration`, which expresses in a finer and more continuous manner the overall active interaction time of the user on the page;
- **BounceRates**: strongly correlated with `ExitRates`, with the latter being preserved by virtue of a higher Mutual Information score;
- **VisitorType_Returning_Visitor**: removed because it is linearly dependent on the other binary variables obtained from the one-hot encoding for the same category.

Thanks to this filtering pipeline, the dimensional space was successfully reduced from 55 to 21 final features and the dataframes called `X_train_sel` and `X_test_sel` were created. This contraction not only mitigates the risk associated with dimensionality, but also ensures greater computational stability for models sensitive to spatial distances or linear coefficients.

5.3 Synthetic Balancing: SMOTE-NC

One of the main obstacles in the dataset is the natural asymmetry in the class distribution, a characteristic of e-commerce data, where sessions ending in an active transaction represent a minority fraction of the total. If ignored, this strong imbalance (approximately 5.5 to 1, fig. 2) leads the estimators to minimize the global loss by simply collapsing onto the most frequent class. Consequently, an apparently high overall accuracy is obtained but with null or almost null predictive power on the class of actual commercial interest, namely the purchasing users.

To mitigate this phenomenon, the SMOTENC technique (*Synthetic Minority Over-sampling Technique for Nominal and Continuous*) was chosen, as extension of the classic SMOTE algorithm designed specifically to handle mixed-

nature datasets, containing both quantitative and categorical variables. Unlike random sampling with replacement, which introduces a high risk of overfitting by slavishly duplicating existing records, SMOTENC generates new synthetic samples by geometrically interpolating the features in the multidimensional space. For continuous variables, the algorithm identifies the k -nearest neighbors within the minority class and generates new points along the segments connecting such neighbors. For categorical variables, the algorithm assigns the most frequent value among the considered neighbors, preserving the logical consistency of the categories and avoiding the creation of improper hybrid combinations or decimal values devoid of physical meaning.

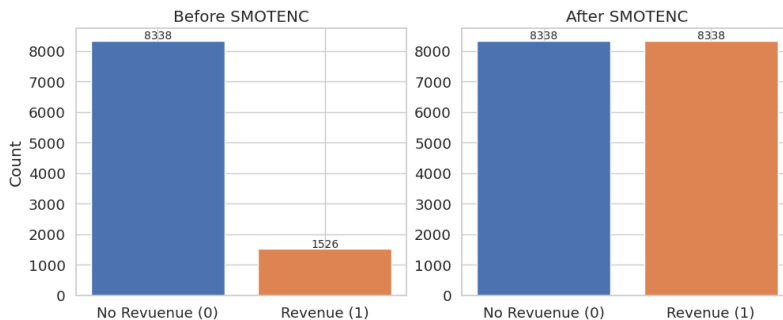


Figure 17: Comparison of the class distribution before and after SMOTENC.

The empirical application of the technique, whose distributional effects are illustrated in Figure 17, highlights a perfect statistical balance between the two classes in the training set. The number of instances associated with the minority class was artificially increased from 1.526 to 8.338, aligning exactly with the cardinality of the majority class and resulting in a final size of the training matrix equal to 16.676 total observations. It is of fundamental importance to emphasize the methodological correctness of this procedure, applied strictly and exclusively on the training set, which became `X_train_res`. The test set was in no way altered by the algorithm, thus ensuring that the final evaluation of the models occurs on real data reflecting the actual imbalanced distribution of the reference e-commerce market.

6 Supervised Learning Modeling

Once the data pre-processing phase is concluded, the supervised modeling phase begins. In the context of the Online Shoppers Purchasing Intention dataset, the main objective consists in correctly classifying the user’s final purchasing intention (`Revenue = True`).

Within the field of algorithms applied to predicting purchasing behavior, the design pipeline identified was developed with the aim of exploring models based on different mathematical and methodological foundations.

Decision tree-based models represent a benchmark for tabular data. The choice of Random Forest is motivated by its nature as an *ensemble* method based on the principle of *bagging*. This approach allows for the combination of a multitude of weak decision trees to obtain a strong classifier, characterized by high statistical stability.

In order to explore supervised learning paradigms alternative to bagging, the predictive architecture was integrated with the XGBoost (eXtreme Gradient Boosting) algorithm, a highly optimized and efficient implementation of the Gradient Boosting framework. Unlike Random Forest, which relies on the parallel and independent training of decision trees to reduce overall variance, XGBoost operates according to a sequential *boosting* principle. In this scheme, individual estimators are constructed sequentially, minimizing a differentiable loss function at each iteration through gradient estimation. Each new tree is then trained specifically to correct the classification errors made by the set of previously generated predictors.

In order to broaden the analytical scope and compare the previous methods with a radically different classification paradigm, the predictive analysis was extended to Support Vector Machines (SVM). Unlike Random Forest and XGBoost, SVMs adopt a purely geometric approach. The algorithm operates by searching for the optimal hyper-plane capable of separating the two classes by maximizing the so-called “margin,” that is, the distance between the decision boundary and the observations closest to it, known as support vectors. By using the *kernel trick*, SVMs can also implicitly project features into a high-dimensional space, allowing for the creation of continuous separation boundaries to model highly nonlinear relationships.

Every model mentioned so far has been tuned by using *Grid Search* coupled with a *Stratified 5-Fold Cross-Validation*:

by defining the grid as a dictionary of possible values for the hyperparameters, several possible combinations were tested; by dividing the test set into five parts, however, we treat one-fifth of the data as a validation fold and calculate the final score as the average of the scores across the folds. Where possible, it was selected the one that maximizes performance; in the event that the empirical results do not clearly indicate the best model, both the baseline and the tuned versions were presented.

Finally, in order to explore deep learning architectures on a typical tabular dataset, the last model used in the framework is the Multi-Layer Perceptron (MLP), a very basic class of feedforward artificial neural networks with multiple hidden layers between the input and output layers.

The performance of these models is compared using evaluation metrics based on the proportion of correct versus incorrect classifications relative to the total number of tests performed or to the correct labels, which are previously known.

The strong asymmetry in the distribution of the target variable requires the adoption of a rigorous, multidimensional evaluation protocol. In scenarios characterized by marked class imbalance, the use of standard aggregate metrics risks being misleading, masking the model's inability to identify purchasing patterns: the basic ratio of correct predictions to the total number of cases reveals a critical structural limitation on imbalanced data, where it can appear deceptively high simply by consistently predicting the majority "non-purchase" class.

It is therefore necessary to define a robust framework of metrics capable of measuring both the statistical robustness of classifiers and their impact on e-commerce business, balancing the cost of missed conversions against that of false positives.

To this aim, the comparative analysis of the models will be conducted on the basis of the following evaluation metrics:

- **Precision**

Fraction of actual purchases among all sessions classified as such; from a business perspective, it is related to the control of false positives, limiting the waste of budget on marketing campaigns targeted at users with no purchase intent.

- **Recall**

Ability of the model to identify actual buyers out of the total of real customers; it reflects the ability not to miss conversions, quantifying the opportunity cost associated with false negatives.

- **F1-Score**

Harmonic mean of Precision and Recall, used as a synthetic and robust indicator to describe the classifier's stability on the purchase class, ensuring that neither of the two metrics is optimized at the expense of the other.

- **ROC-AUC** (*Area Under the Receiver Operating Characteristic Curve*)

Global measure of the model's discriminative ability between sessions with and without purchase as the classification threshold varies, computed as the area under the curve of the true positive rate against the false positive rate.

- **PR-AUC** (*Area Under the Precision-Recall Curve*)

It represents the metric of choice for imbalanced datasets since it focuses exclusively on the predictive quality of the minority class (**Revenue = True**), isolating the effect of the numerous true negatives.

Another key concept in classification performance is the confusion matrix, a contingency table that systematically compares the actual classes with the predicted ones: one dimension of the matrix represents the instances in a predicted class, while the other represents the instances in an actual class. This tool provides a visual representation of how the model's errors are distributed between false positives and false negatives.

The evaluation of the implemented classifiers on the test set is carried out by the function `evaluate_and_log_model` defined in block 4.2, while a graphical content of the obtained results is provided by `plot_model_dashboard` defined later in the same block.

6.1 Random Forest

The intrinsic robustness of the Random Forest against overfitting is ensured by the decorrelation of the trees, achieved not only through bootstrap sampling of the training sessions, but above all through Feature Bagging (in

the case under examination by setting `max_features = 'sqrt'`), which forces each split to evaluate only a random subset of variables. This property, together with the ability to capture complex non-linear interactions, makes the algorithm ideal for modeling the complex navigation of e-commerce users without requiring restrictive assumptions on the data distribution.

The study was structured into two sequential phases.

First, in block 4.3, a robust and computationally efficient baseline benchmark was defined, introducing constraints on maximum depth (`max_depth = 12`) and minimum leaf cardinality (`min_samples_leaf = 10`).

Subsequently, in block 4.4, the model was implemented by varying within the hyperparameter space through Grid Search and Stratified 5-Fold Cross-Validation. This step aims to mitigate the proverbial trade-off between bias and variance, optimizing sensitivity on the minority class and maximizing the generalization capability.

The different choices are shown in table 11.

Hyperparameter	RF Baseline	RF Tuned
<code>n_estimators</code>	500	500
<code>max_depth</code>	12	None (Senza limite)
<code>min_samples_leaf</code>	10	5
<code>max_features</code>	sqrt	sqrt
Internal Validation method	OOB Score	5-fold Cross-Validation

Table 11: Hyperparameters comparison: RF Baseline vs RF Tuned (best in Grid Search).

Out-of-Bag (OOB) Score is an internal validation metric specific to the Random Forest algorithm that, by utilizing the samples excluded from the *bootstrap* of each tree, provides an unbiased estimate of the generalization error without requiring a validation set.

In the crucial evaluation phase, in addition to measuring the individual performances, one is confronted with the methodological dilemma of evaluating whether the substantial computational effort of hyperparameter tuning (approximately 16 minutes of grid search coupled with a 5-fold Cross-Validation, for a total of 180 independent fittings) has unlocked a significant advantage over the RF model baseline.

For a visual support to the diagnosis of the quality returned by the models, the evaluation plots on the test set are introduced (fig. 18 and fig. 19).

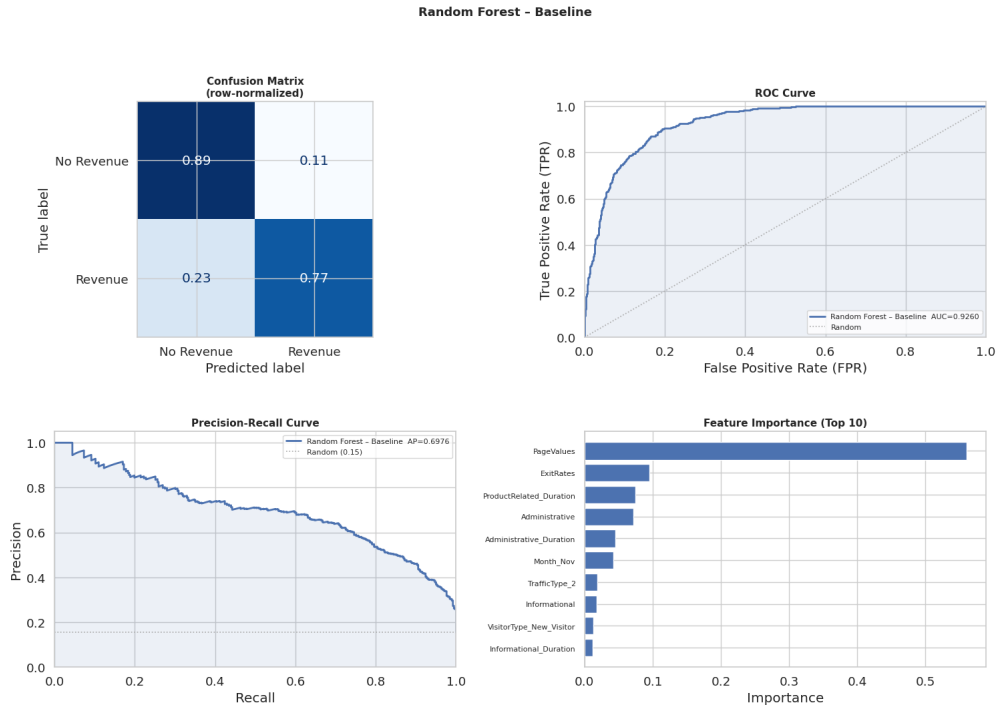


Figure 18: Performance diagnostics of the baseline RF model on the Test Set.

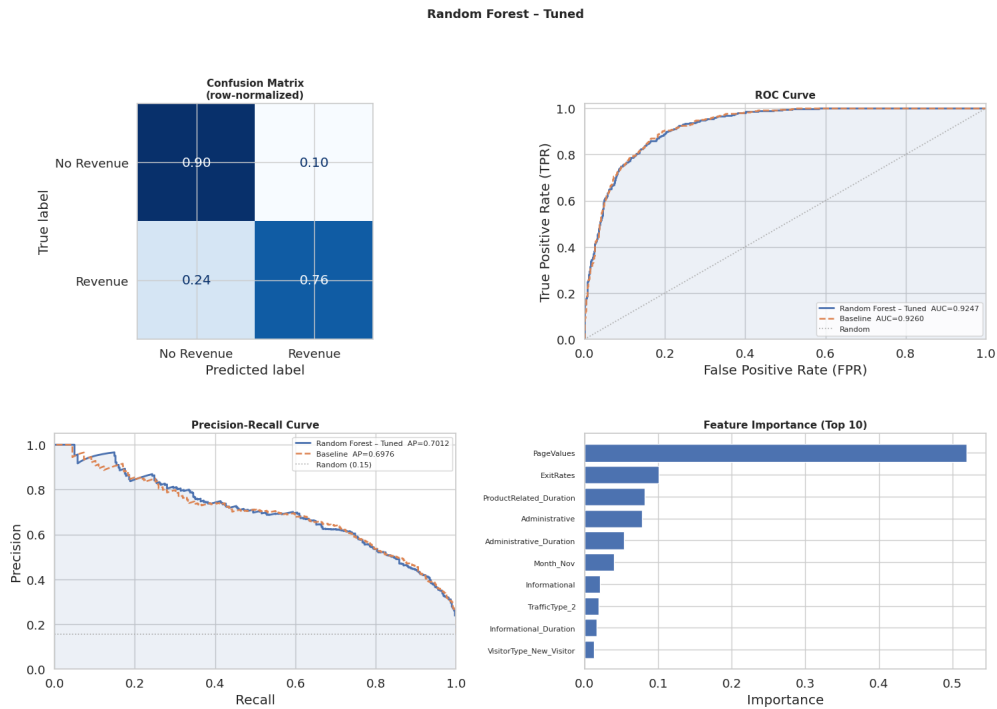


Figure 19: Performance diagnostics of the tuned RF model on the Test Set with ROC/PR-AUC comparison.

To clearly perceive the effect of the tuning, table 12 summarizes the metrics obtained from the test set, highlighting the Δ induced by the hyperparameter optimization.

Metric	RF Baseline	RF Tuned	Δ (Tuned - Baseline)
ROC-AUC	0.9260	0.9247	-0.0013
PR-AUC (AP)	0.6976	0.7012	+0.0036
Precision (Classe 0)	95%	95%	0%
Recall (Classe 0)	89%	90%	+1%
F1-Score (Classe 0)	92%	92%	0%
Precision (Classe 1)	57%	58%	+1%
Recall (Classe 1)	77%	76%	-1%
F1-Score (Classe 1)	66%	66%	0%

Table 12: Performance comparison on the Test Set: Baseline vs. Tuned Random Forest

The comparative analysis of the data reveals that the tuning process acted predominantly as a structural regularization operator, inducing a millimetric variation in the model’s decision boundary.

The optimized model records a very small increase in Precision for the conversion class (from 0.57 to 0.58), counterbalanced by a specular loss in Recall (from 0.77 to 0.76).

From an economic-business perspective, this change describes a more selective classifier: the increase in precision reduces background noise and the waste of advertising budget due to False Positives (erroneously labeled as buyers), while accepting the omission of an infinitesimal fraction of real transactions.

In the presence of the pronounced class imbalance described in the previous chapter, the stability of the area under the ROC curve (which undergoes an irrelevant contraction of -0.0013) proves to be a partially optimistic and misleading metric. Conversely, the area under the Precision-Recall curve, which rises from 0.6976 to 0.7012 ($+0.0036$), provides the mathematical proof of the actual qualitative improvement of the model tuned on the minority class **Revenue = True**.

The maintenance of an F1-Score equal to 66% for both models attests to the symmetric and conservative nature of the calibration. The tuning simply repositioned the implicit decision threshold without altering the overall expressive capacity of the estimator.

From a theoretical-interpretative perspective, the extraction of Feature Importance based on the Gini impurity index repropose the already known highly polarized behavior, with **PageValues** dominating the overall architecture with 51.96% of the importance in the optimized model and even 56.07% in the baseline configuration.

Although the high informative capacity of **PageValues** is consistent with the business logic of e-commerce, such a marked concentration of importance introduces a severe structural vulnerability. In case technical anomalies, latencies, or tracking errors occur in the Google Analytics monitoring pixels, the predictive reliability of the entire Machine Learning pipeline would undergo a sudden degradation. The tuning of the hyperparameters (in particular the selection of **min_samples_leaf = 5**) has partially mitigated this dependence, redistributing approximately 4% of importance to subsidiary behavioral variables such as **ExitRates** (10.09%) and **ProductRelated_Duration** (8.17%).

In general, the scarce performance oscillations recorded upon the Grid Search mathematically indicate that the geometric space generated by the union of SMOTENC and RF Baseline was already in extreme proximity to the global optimum, downgrading them to mere statistical fluctuations and sampling noise.

Therefore, the final algorithmic decision ceases to be a purely mathematical problem to become a strategic business choice.

Two specular scenarios are identified:

1. In the case of an e-commerce operating under stringent budget constraints, the use of the Tuned version of RF is recommended. In this scenario, the maximization of the PR-AUC and the gain in Precision ensure the reduction of costs associated with false alarms, maximizing the return on advertising investment.
2. If the IT architecture requires maximum computational efficiency or frequent real-time model updates, the baseline configuration is preferred. With an equal F1-Score, the baseline nullifies the technical and energy expenditure related to the periodic re-execution of the Grid Search, while simultaneously offering a superior Recall. In high-margin markets, where the omission of a potential buyer represents a loss, this choice proves ideal.

6.2 XGBoost

The implementation of XGBoost introduces significant structural improvements compared to Random Forest. From a mathematical standpoint, this new framework incorporates a formal regularization term (L_1 and L_2) directly into the objective function, actively limiting the complexity of the trees to prevent overfitting phenomena (RF relies almost exclusively on the random sampling of features and observations to ensure generalization). Furthermore, while Random Forest tends to be affected by the intrinsic noise in heavily imbalanced datasets due to its non-sequential nature, XGBoost proves inherently more capable of capturing non-linear decision boundaries, dynamically weighting the instances that are most difficult to classify.

In the present scenario of strongly asymmetric classification, the training was conducted by exploiting the space of 21 pre-processed features balanced with SMOTENC (`X_train_res`).

Comparing the test set results obtained in blocks 4.5 and 4.6, the baseline model was chosen, as showed the best performance among the computed versions.

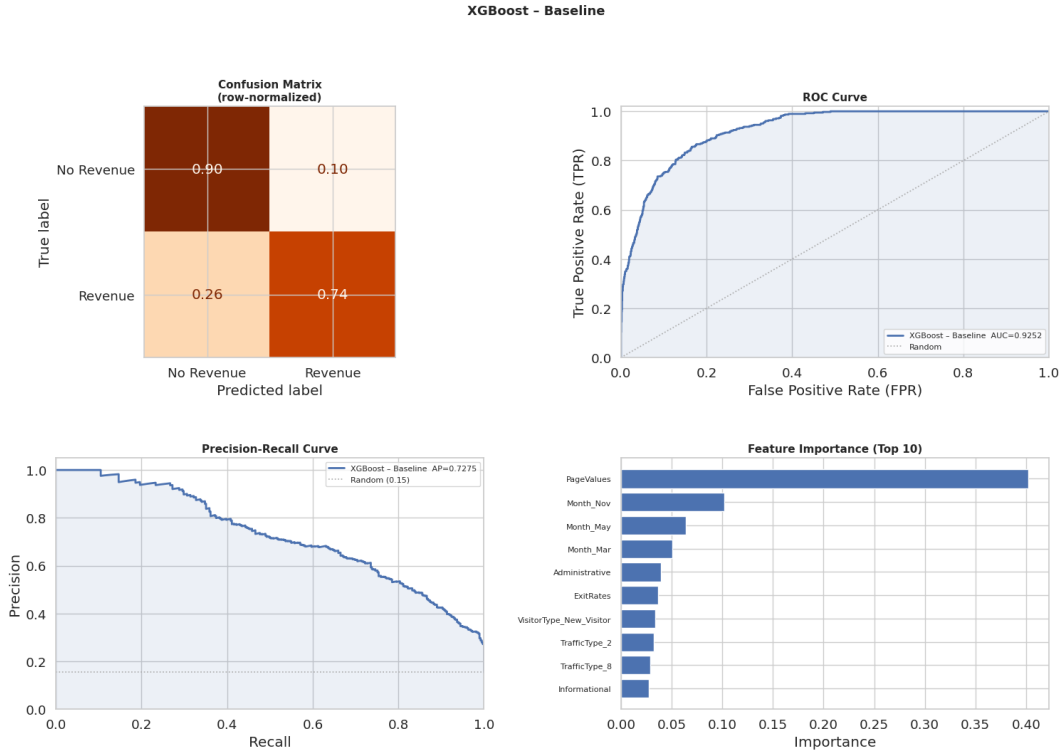


Figure 20: Performance diagnostics of the XGBoost model on the Test Set

The quantitative results recorded on the independent test set are reported in Table 13.

Classe	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Revenue = 0	95%	90%	93%	0.9252	0.7275
Revenue = 1	59%	74%	65%		

Table 13: Performance metrics on the Test Set for the XGBoost Baseline model.

Focusing the analysis on the conversion class, the classifier exhibits a Precision of 59% associated with a Recall of 74%, synthesized into an F1-Score equal to 65%.

In a business context, a higher precision value reduces the waste of advertising budget, ensuring that almost 60% of the users targeted by real-time promotional campaigns (such as push notifications or personalized discounts at checkout) will more probably finalize the transaction.

On the other hand, the model accepts a marginal increase in False Negatives (equal to 26%), allowing a share of real buyers to conclude the session without being intercepted by the portal’s stimulus actions.

The visual analysis of the curves and the breakdown of errors (fig. 20) provides crucial empirical evidence on the model’s generalization capability.

The matrix shows an excellent ability to isolate the majority class, with 90% of true negatives correctly identified and an error rate limited to 10%. For the same class, the decomposition highlights how 74% of the converted sessions are successfully captured, against a 26% of predictive omissions. This confirms that the model does not suffer from bias toward the prevailing class despite the asymmetry of the original dataset.

The ROC plot highlights a high overall discrimination. The curve rises sharply toward the upper-left corner, demonstrating that the model maintains an extremely high true positive rate against very contained false positive rates. The subsequent stabilization of the curve at a value of 0.93 certifies the model’s effectiveness in correctly ranking the confidence level associated with random pairs of customers.

The PR curve represents the most stringent and informative metric for imbalanced scenarios.

The Area Under the Curve value (PR-AUC/AP) stands at 0.73, clearly exceeding the random prevalence baseline set at 15% (which is the volume of the positive class, standardly used as reference). The trend of the curve shows remarkable stability: the *Precision* remains above 80% for *Recall* levels up to 40%, before degrading in an almost always controlled manner. This property ensures that, if the e-commerce decide to operate with very high confidence thresholds for campaigns with very high transactional cost, the classifier would maintain an exceptional targeting effectiveness.

6.3 Support Vector Machines (SVM)

The methodological transition from tree-based models to those based on margin maximization introduces substantial differences in the handling of information. While decision trees excel at the logical and hierarchical decomposition of the original variables, SVMs are distinguished by their strong mathematical regularization and by their exclusive focus on "borderline" cases. Since the cost function of SVMs ignores the observations already correctly classified with a wide margin, the model is inherently less inclined to memorize the global density of the noise, concentrating learning solely on the most ambiguous and hard-to-discriminate browsing sessions. The introduction of this algorithm therefore makes it possible to verify whether users’ purchase intention is more easily isolable through a spatial partitioning of the multidimensional plane rather than through cascading conditional rules.

Due to the highly clustered nature of the data, the *kernel trick* was used to overcome linear inseparability: it projects the original data points into a higher-dimensional space in order to make them linearly separable in that space. The transformation does not occur explicitly, but the kernel SVM computes the decision boundary in terms of similarity measures in a high-dimensional feature space without actually performing the projection. In this case, this similarity was calculated using *Radial Basis Functions* (RBF) in the corresponding RBF kernel.

In the present study, basing on the performances resulted from the Grid Search and CV, the chosen classifier was the baseline SVM RBF. As shown in the output of the block 4.7, its training culminated in the identification of 6000 support vectors, equal to 36% of the entire training set.

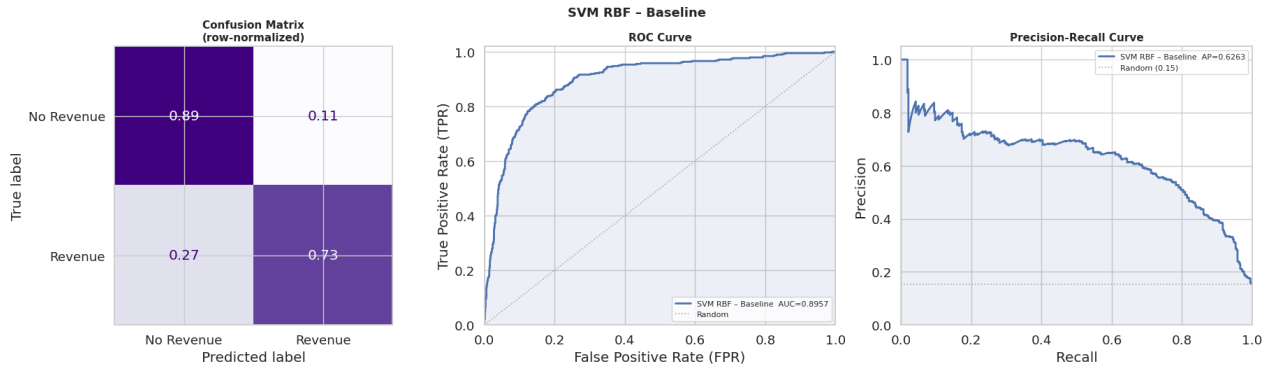


Figure 21: Performance diagnostics of the SVM model on the Test Set

The quantitative results recorded on the test set are presented in Table 14.

Classe	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Revenue = 0	95%	89%	92%	0.8957	0.6263
Revenue = 1	56%	73%	63%		

Table 14: Performance metrics on the Test Set for the SVM RBF Baseline model.

On the **Revenue = True** class, the implemented SVM model exhibits a Precision of 56% and a Recall of 73%, with a resulting F1-Score of 63%. From the perspective of online business, such a Precision implies that approximately 44% of the users classified as "purchase-intending" will actually abandon the session without completing the transaction.

The visual decomposition provided by the confusion matrix (table 14) offers a deeper interpretive key regarding the intrinsic quality of the geometric separation achieved. The model correctly classifies 89% of the non-finalized sessions, allowing only 11% of the **Revenue = False** predictions to generate a false alarm. On the positive class, the proportion of true positives stands at 73%, against an omission rate equal to 27%.

For the first time among the analyzed models, the area under the ROC curve drops below the 0.90 threshold, settling at 0.8957.

The analysis of the Precision-Recall curve highlights the most critical issues. The PR-AUC value at 0.6263, although it widely surpasses the random prevalence line of 15%, reflects a marked degradation of Precision as Recall increases. In particular, the trend of the curve shows strong oscillations and a rapid decay of precision for recall values above 60%. This unstable behavior denotes a less refined calibration of the output probabilities, making the SVM RBF model poorly suited to operate in flexible application scenarios where the decision threshold must be dynamically recalibrated.

The theoretical explanation for this performance gap lies in the very nature of the "Online Shoppers Purchasing Intention" dataset. The variables governing purchase intention exhibit strongly asymmetric distributions, non-uniform scales, and complex nonlinear interactive effects (consider the correlation between **PageValues**, **ExitRates**, and session duration).

6.4 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) represents a fundamental class of artificial neural networks. From a structural standpoint, the architecture is organized into at least three levels of nodes: an input layer, one or more hidden layers, and an output layer. With the exception of the input units, each node applies a nonlinear activation function to map the linear combinations of the incoming signals. In the specific case, nonlinearity was handled through the ReLU (*Rectified Linear Unit*) function. The training process is based on the *backpropagation* algorithm, which iteratively updates the synaptic weights via gradient descent in order to minimize the loss function.

In order to evaluate the impact of architectural complexity and regularization strategies on performance, two distinct configurations were employed: a Baseline model and a 2 Hidden Layers (2HL) model.

While the Baseline relies on the standard single hidden layer architecture, the 2HL version was designed with a two hidden layer architecture of 64 and 32 neurons. This choice aims to force the network to extract increasingly compressed and high-level latent representations.

To mitigate the propensity for overfitting, the 2HL model integrates two control mechanisms: an increase in the L2 penalization coefficient on the weights and the activation of Early Stopping. The latter dynamically halts training if the loss computed on an internal validation set (equal to 10% of the training pool) does not show improvements for a certain epoch tolerance, the patience (set to 15 consecutive epochs in the code), ensuring a data-driven optimization and preventing the memorization of noise.

A summary of the two perceptron configurations implemented in blocks 4.9 and 4.10 is provided below.

Hyperparameter	MLP Baseline	MLP 2HL
Architecture (Hidden Layers)	(100,)	(64, 32)
Activation Function	ReLU	ReLU
Optimization Algorithm	Adam	Adam
Learning Rate (LR)	Constant	Adaptive
L2 Penalization (α)	0.0001	0.01
Early Stopping	False	True
Validation Fraction	-	10%
Patience	-	15
Effective Epochs	448	113

Table 15: Hyperparameters and convergence comparison: MLP Baseline vs MLP 2HL

The following tables summarize the performance metrics on the test set for both computed configurations.

Metrica	MLP Baseline	MLP 2HL	Δ (2HL - Baseline)
ROC-AUC	0.8913	0.8865	-0.0048
PR-AUC (AP)	0.6498	0.6111	-0.0388
Precision (Classe 0)	94%	94%	0%
Recall (Classe 0)	88%	85%	-3%
F1-Score (Classe 0)	91%	90%	-1%
Precision (Classe 1)	52%	48%	-4%
Recall (Classe 1)	68%	73%	+5%
F1-Score (Classe 1)	59%	58%	-1%

Table 16: Performance comparison on the Test Set: Baseline vs. 2HL

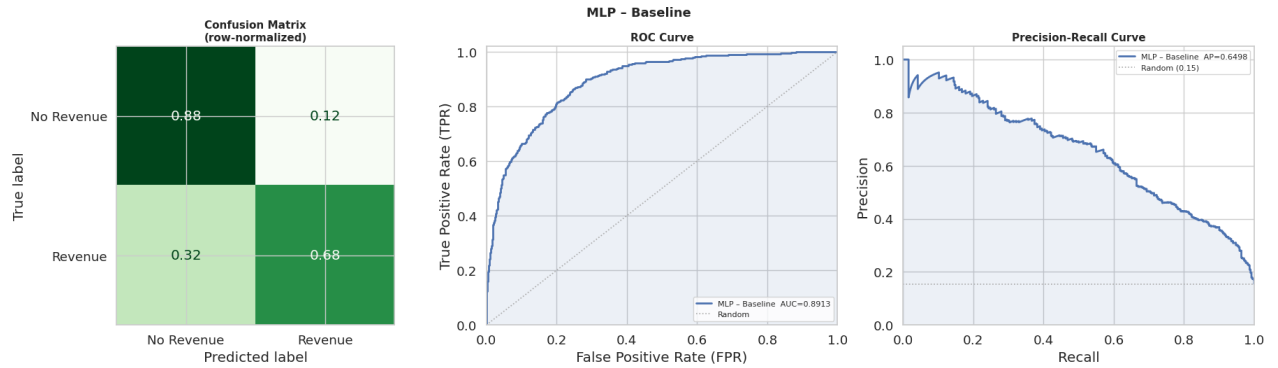


Figure 22: Performance diagnostics of the baseline MLP model on the Test Set

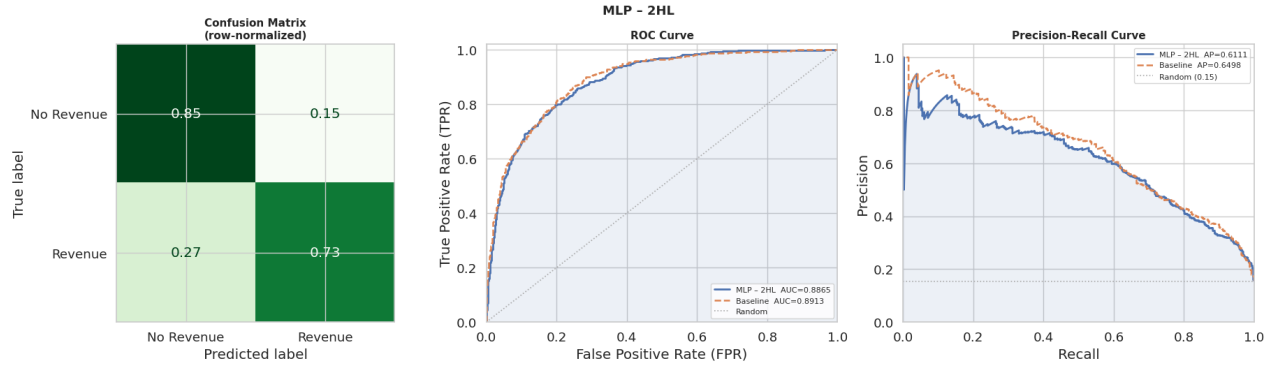


Figure 23: Performance diagnostics of the 2 Hidden Layers MLP model on the Test Set with ROC/PR-AUC comparison.

As the data shows in table 16, the new configuration did not yield an absolute improvement in the global metrics, revealing instead a decrease in the PR-AUC (from 0.6498 to 0.6111). The more complex architecture of the 2HL model entailed a marked trade-off between Precision and Recall on the target class.

The baseline model maintains a more conservative and balanced behavior, ensuring a lower False Positive rate with a Precision of 52%.

The 2HL model, on the contrary, heavily sacrifices Precision (48%) to maximize the ability to intercept buyers, pushing the Recall to 73% and returning an error rate of 15% on the negative class (compared to 12% for the Baseline).

A visual confirmation of these dynamics is provided by the graphical analysis of the ROC and Precision-Recall curves. While the ROC curves of both models exhibit comparable convexity, with the 2HL trajectory (solid line in fig. 23) sitting marginally below the Baseline (dashed line) across almost all False Positive Rate thresholds, the decisive discriminative discrepancy emerges in the PR space.

Observing the Precision-Recall curve of the 2HL model, one can notice a much sharper structural drop in Precision even at low Recall thresholds. Unlike the Baseline, which maintains a relatively stable Precision plateau up to a Recall of 0.4, the 2HL curve drops abruptly. This graphical behavior perfectly mirrors the row-normalized confusion matrices: the more complex architecture expands its decision boundary to capture more buyers (73% vs 68%), but in doing so, it severely dilutes the predictive confidence, pulling down the Precision curve much earlier due to the immediate influx of False Positives.

The inclusion of both results is fundamental for the critical analysis. Evaluated within the neural ecosystem alone, the choice between the two models is not absolute, but depends on the business objective: the Baseline model is to be preferred to maximize the Return on Investment in costly marketing campaigns, avoiding waste on False Positives; the 2HL model would be useful only in very low-cost retargeting scenarios where the priority objective is not to lose potential sales, prioritizing a high Recall.

7 Comparisons and Conclusions

The evaluation of six distinct machine learning architectures provides a comprehensive overview of the predictive capabilities on the Online Shoppers Purchasing Intention dataset. The primary objective was to isolate the optimal algorithm capable of predicting the **Revenue** class while balancing the trade-off between Precision and Recall, a critical requirement for targeted e-commerce marketing strategies.

With this in mind, comparative charts were created in sections 5.1 and 5.2 to provide a clear visual overview of the performance results. As the primary class of interest in the e-commerce business is the one that is most difficult to predict due to its significant imbalance, the focus has been on the **Revenue = 1** class.

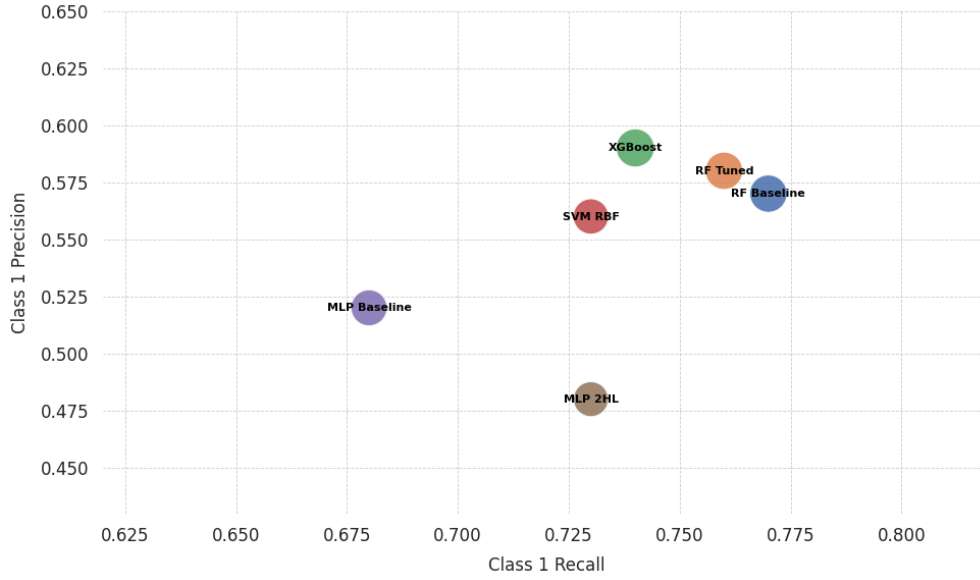


Figure 24: Comparative Scatterplot – Precision vs Recall (Class 1)

The scatterplot in fig. 24 plots each model as a single bubble in the two-dimensional Recall vs Precision space, and the size of each bubble is proportional to the PR-AUC. This allows you to instantly assess the trade-off between false positives and false negatives on the minority class. The higher and farther to the right a model is positioned, the more effective it is at maximizing actual conversions while reducing false alarms; meanwhile, the size of the bubble visually highlights the model’s overall robustness under class imbalance.

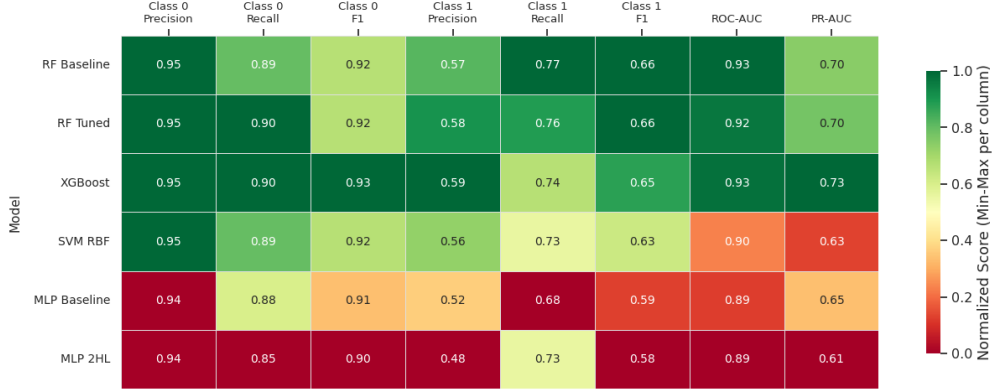


Figure 25: Performance Comparison Heatmap – Online Shoppers Purchasing Intention

The heatmap in fig. 25 displays the models in the rows and all performance metrics in the columns. The color gradient is calculated by applying *Min-Max* normalization independently to each column, although the cells show the actual performance values. Column normalization allows for an immediate relative comparison: the viewer can scroll vertically through a metric and instantly identify, based on color intensity, which model excels or falls short on that particular indicator.

Empirical analysis of the results on the test set reveals that decision tree-based architectures (Random Forest and XGBoost) are clearly superior to models based on distances (SVM) or neural networks (MLP), as notable from the position of the bubbles in the scatterplot and from the stark green predominance in the upper rows of the heatmap. This difference is not coincidental, but stems directly from the mathematical foundations of the algorithms when applied to heterogeneous and heavily imbalanced tabular datasets. The RBF SVM and MLP models confirm an inherent difficulty in handling highly heterogeneous web data, consisting of continuous variables with high variance mixed with binary and sparse categories. Although the SVM’s intrinsic geometric regularisation protects it from

a complete predictive collapse towards the majority class, the decline in ROC-AUC (0.8957 compared with 0.9252 for XGBoost) is explanatory. Whilst trees capture local interactions by recursively partitioning the space, the RBF SVM attempts to map relationships globally using Euclidean distance functions. The extraction of a very large number of support vectors (equivalent to 36% of the training set) indicates that the decision boundary identified is extremely jagged: a clear sign of the model’s difficulty in generalising in a linear and clean manner, leading to local micro-overfitting. Similarly, the MLP models (Baseline and 2HL), whilst offering excellent insights into predictive trade-offs, fail to exceed the 0.65 threshold for PR-AUC (visibly highlighted in the warm tones in the heatmap), suffering from their well-known complexity in tuning on imbalanced tabular data.

Although Random Forest and XGBoost dominate the overall metrics, they exhibit profoundly different operational behaviours. As it could be noted from fig. 18, fig. 19 and fig. 20, Random Forest (both Baseline and Tuned) maximises Recall (standing on the furthest right of the scatterplot), but exhibits a marked ‘laziness’ in feature extraction: mathematically speaking, independent sampling drives individual trees to greedily select the single variable with the greatest discriminatory power in order to reduce impurity. In contrast, XGBoost, thanks to its sequential, gradient-based nature, focuses on the residuals left unexplained by previous iterations. Visually, this translates to occupying the highest position on the Y-axis, ensuring a more balanced representation and better control over false positives. The feature importance analysis in Random Forest models revealed a marked polarisation of predictive power, with the single metric **PageValues** accounting for over 50% of the total importance. This exposes the model to a potential hierarchical dependency. In a production environment, this architecture is inherently fragile: if the tracking of the **PageValues** feature experiences anomalies, delays or corruption in the real-time stream, the model would lose its primary basis for decision-making, resulting in a drastic drop in performance. By contrast, XGBoost demonstrates greater resilience. By providing a much more robust representation of the feature space, it mitigates dependence on a single predictor, facilitates greater control over false positives and delivers the best PR-AUC value observed (0.7275, the darkest green in the respective heatmap column).

Consequently, XGBoost is elected as the final model for this project: by coupling the highest precision-recall balance on the minority class with a resilient, diversified feature utilization, it provides the most robust and deployment-ready solution for predicting online purchasing intentions while minimizing the operational costs of false positive alarms.

8 AI Disclosure

During the development of the code and the writing of this paper for the Online Shoppers Purchasing Intention Dataset analysis, generative AI tools were used for the following purposes:

- support for the initial brainstorming aimed at identifying the project pipeline,
- support for code debugging,
- provision of internal documentation within the code,
- improvement of code readability,
- improvement of the expository flow during writing where deemed appropriate.

The models consulted are listed below:

- Claude Opus 4.8 by Anthropic,
- Claude Sonnet 4.6 by Anthropic,
- Gemini 3.1 Pro by Google,
- Gemini 3.5 Flash by Google.

After consulting these tools, the generated content was always reviewed, edited, and validated with great care and critical thinking. The methodological choices, the selection of hyperparameters, the interpretation of the results, the writing of the content and the formulation of the final conclusions are the exclusive result of human reasoning and analysis.

9 References

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